



Course syllabus

CMG 1000 - Introduction to Construction Management

Term: Fall 2025

Course Time: Monday, Wednesday and Friday: 9:00 AM to 9:50 AM

Classroom: Klehm Hall 2040

Course Dates: Aug. 25, 2025 – Dec. 12, 2025

Faculty: Dr. Reachsak Ly

Email: rly@eiu.edu (Preferred communication method)

Office: Klehm Hall 4014

Teaching Method: Lecture

Office Hours: Tuesday and Thursday, 11:00 AM to 1:00 PM or by appointment

Communication Response Time: Within 2 days

Required Textbooks and Readings

- **Required Textbook:** Jackson, B.J. (2020). Construction management jumpstart: The best first step toward a career in construction management. 3rd edition. Wiley & Sons, Hoboken, New Jersey
- **Suggested Textbooks:** Construction Project Management 2nd Edition by Dykstra published by Kirshner ISBN 9780-9827034-3-4
- **Readings:** Several readings will be posted on D2L.

Course description:

Major topics that will be discussed in class will include: The Construction Industry and Trends, Design and Construction Project Phases, Construction Contracts, Project Management, Pre-Construction, Estimating, Scheduling, Project Controls, Project Administration, Quality Control, Health and Safety, BIM, Sustainability and Technologies application in construction.

Course objectives:

This course will introduce the students to construction management processes and procedures, relationships, practices, terminology, project types, procurement methods, industry standards, contract documents, and career opportunities. Topics include the roles of the owner, architect, engineer, and contractor; project teams and organizations; their responsibilities and interrelationships; types of contracts; different project delivery methods; bidding and award documents and procedures; construction budgets; cost estimating; construction planning and scheduling; cost control; sustainability; technology; and managing quality, health and safety. In addition, the course will introduced different technologies used in construction industry.

A tentative schedule for this course is attached in the Course outline section below.

Student Learning outcome:

After completion of the course the student will:

- Understand the characteristics and challenges of the construction industry within the U.S. and role of construction manager/responsibilities.

- Understand the concepts relating to construction administration and project delivery systems, contracting requirements, project control and tracking, dispute resolution, safety, quality assurance and control, and bid process to contract award.
- Understand the principles of project scheduling/stages and explain network diagramming process leading to critical path method and calculations.
- Understand the basic building construction estimates for time, labor, and materials.
- Understand the latest technologies used in the construction industry.

Course Structure

- This course is mainly lectures and projects. You are required to attend each lecture, participate in the class discussion, and actively contribute to your group project.
- This course includes several homework assignments and pop-up quizzes.
- Two exams are scheduled for this course and NO final exam for this course.
- One team project is included for this course.

Graded Course Requirements Information

Homework Assignments and Quizzes

- The homework assignments cover all contents from lectures, textbooks, readings and class discussion.
- The quizzes are unannounced. Please note, there will be no chances for a pop-up quiz make up.
- Attending all lectures is the only way to not miss a pop-up quiz.

Team Projects

The team project includes two deliverables: (1) in-class presentation, (2) a report summarizing your findings written in an academic manner. Detailed information will be discussed in the lectures.

Exams

Two exams cover lectures, textbook, readings and class discussion.
No final exam is scheduled for this class.

Grading Structure

Content	Percentage
Attendance and Participation	10%
Homework assignment and quizzes	25%
Team Project	15%
Exam 1	25%
Exam 2	25%
Total	100%

Grading Scale:

Individual grade items and your final grade will be assessed as follows:

A = 100- 90%

B = 89- 80%

C = 79-70%

D = 69-60%

F = less than 60%

Grading Policies

Deductions to your assignments grade for late submission will be given as follows:

0-24 hours late a deduction of 15% of the earned grade

24-48 hours late a deduction of 40% of the earned grade

48-72 hours late a deduction of 60% of the earned grade

More than 72 hours late No credit.

Assignments must still be submitted to get a grade for this course.

Course Requirements:

1. Complete readings, assignments, attend class, and submit assignments and exams on time.
2. Demonstrating that you are interested and enthusiastic about learning is a good way to improve Participation scores. Be present in both body *and* mind.
3. Independent study of the textbook and other relevant material is required.
4. Attendance: If you do not attend and participate you will not succeed in this class. Only **documented** emergencies and legitimate University activity-related events (e.g., athletes for games, sponsored field trips for other classes) will be excused. Tardiness reduces points awarded by ½. No points are awarded for absences.
5. Homework Assignments, Tests, and Communication will be conducted through the campus learning management system. The student is required to access the system *daily* for updates and to submit assignments.

Course outline

* The course schedule is subject to change throughout the semester. Revisions will be noted in class.

Week	Date	Topics	Readings / Assignments / Quizzes
1	25-Aug	Session 1A: Course introduction	Read Syllabus & Chapter 1 and 2
	27-Aug	Session 1B: Construction Industry, Project Players, Construction Management	
	29-Aug	Session 1C: Presentation from the Lumpkin College Hub for Engagement, Leadership and Professionalism	
2	1-Sep	No Class	
	3-Sep	Session 2A: Project Delivery Methods - DBB, DB & CM Agency	Read Chapter 2
	5-Sep	Session 2B: Project Delivery Methods - DB Bridging, CM@Risk, & IPD	
3	8-Sep	Session 3A: Getting the Work	Read Chapter 3
	10-Sep	Session 3B: Project Stage - Predesign & Design Stage	Read Chapter 5
	12-Sep	Session 3C: Project Stage - Construction & Post-Construction	
4	15-Sep	Session 4A: Managing Teams	Read Chapter 6
	17-Sep	Session 4B: Contracts – Drawings and Spec	Read Chapter 4

	19-Sep	Session 4C: Contracts - Contract Types	
5	22-Sep	Exam review	
	24-Sep	Exam 1	
	26-Sep	Session 5A: Preconstruction	Read Chapter 7
	29-Sep	Session 6A: Estimating	Read Chapter 8
6	1-Oct	Session 6B: Contract Administration	Read Chapter 9
	3-Oct	Session 6C: Change Orders	
	6-Oct	Session 7A: Construction Operation	Read Chapter 10
7	8-Oct	Session 7B: Site Management	
	10-Oct	Session 7C: Planning and Scheduling 1	Read Chapter 11,
8	13-Oct	Session 7C: Planning and Scheduling 2	Read Chapter 11,
	15-Oct	Session 8A: Managing Quality & Safety	Read Chapter 13
	17-Oct	No Class	
	20-Oct	Workshop #1	
9	22-Oct	Session 8B: Monitoring Project Performance	Read Chapter 12
	24-Oct	Session 8B: Value Engineering and EVM	
	27-Oct	Session 9A: Risk Management	Read Chapter 14
10	29-Oct	Session 9B: BIM	Read Chapter 15
	31-Oct	Session 10A: IoT and Sensing	
	3-Nov	Workshop #2	
11	5-Nov	Session 11A: UAS	
	7-Nov	Session 11B: AR/VR	
	10-Nov	Session 12A: Digital Twin	
12	12-Nov	Session 12B: Machine Learning	
	14-Nov	Workshop #3	
	17-Nov	Session 13A: Sustainability and Built Environment	
13	19-Nov	Session 13A: Lean Construction	Read Chapter 16
	21-Nov	Session 13B: Introduction to LEED	
14		Thanksgiving Break	
15	1-Dec	Exam Review	
	3-Dec	Exam 2	
	5-Dec	Presentation Preparation	
16	8-Dec	Team Presentation	
	10-Dec		
	12-Dec		

Note: This is a tentative schedule, and subject to change as necessary. The instructor reserves the right to make changes to the topics or sequence of topics, schedule, assignments, grading plan, and content of the course.

Academic Integrity:

Students are expected to maintain the principles of academic integrity and conduct as defined in EIU's Code of Conduct (<http://www.eiu.edu/judicial/studentconductcode.php>). Violations will be reported to the Office of Student Standards. In this class, students may assist each other in accomplishing a homework assignment, but the submitted work must be the student's own original work. **Copying work or plagiarism will not be tolerated. Students may *not* collaborate on tests or quizzes. Cheating or plagiarism will, at least, result in a zero on the assignment or exam.**

- *AI and Academic Integrity:* Students are not allowed to use advanced automated tools (artificial intelligence or machine learning tools such as ChatGPT, Grammarly, or Dall-E 2) on assignments in this course. Each student is expected to complete each assignment without substantive assistance from others, including automated tools.

Students with Disabilities:

If you are a student with a documented disability in need of accommodations to fully participate in this class, please contact the Office of Student Disability Services (OSDS). All accommodations must be approved through OSDS. Please stop by Ninth Street Hall, Room 2006, or call 217-581-6583 to make an appointment

Emergency Preparedness Plan:

Instructions regarding procedures to follow in the event of an emergency are posted in all classrooms. Students are responsible for acquainting themselves with these instructions so that they are prepared in the event of an emergency.

The Academic Success Center:

Students who are having difficulty achieving their academic goals are encouraged to contact the Academic Success Center (www.eiu.edu/~success) for assistance with time management, test taking, note taking, avoiding procrastination, setting goals, and other skills to support academic achievement. The Center provides individualized consultations. To make an appointment, call 217-581-6696 or go to 9th Street Hall, Room 1302.

Course syllabus

CMG 2223 - Print Reading and Introduction to Building Information Management (BIM)

Term: Fall 2025

Course Time: Tuesday and Thursday: 1:00 PM to 2:40 PM

Classroom: Klehm Hall 3135

Course Dates: Aug. 25, 2025 – Dec. 12, 2025

Faculty: Dr. Reachsak Ly

Email: rly@eiu.edu (Preferred communication method)

Office: Klehm Hall 4014

Office Hours: Tuesday and Thursday, 11:00 AM to 1:00 PM or by appointment

Communication Response Time: Within 2 days

Required Textbooks and Readings

Required Textbook:

- Print Reading for Construction (6th Edition), Walter C. Brown & Daniel P. Dorfmueller, ISBN: 9781605258027
- Daniel John Stine, "Design Integration Using Autodesk Revit 2022," SDC Publications
- Readings: Several readings will be posted on D2L.

Course Prerequisite:

EGT 2043 - Computer-Aided Engineering Drawing

Course description:

This course provides an introduction to reading and interpreting construction drawings and specifications, along with an entry-level exploration of Building Information Modeling (BIM).

In the first portion of the course, students will learn how construction plans are organized and drawn, how to read and interpret architectural and engineering drawings, and how to apply this knowledge from conceptual design through final construction documentation.

In the second part of the course, students are introduced to the fundamentals of BIM, with a focus on creating and modifying 3D building models using Autodesk Revit. Students will learn how BIM supports the design, visualization, and documentation process in the construction industry, and how it differs from traditional computer-aided design (CAD) methods.

Course objectives:

Upon successful completion of this course, students will be able to:

- Read and interpret dimensions, line types, legends, and schedules.
- Explain the role of specifications and their relationship to drawings.
- Recognize and interpret site plans, floor plans, elevations, sections, and detail drawings.

- Identify symbols and notations for structural, mechanical, electrical, and plumbing systems.
- Interpret construction details for materials and systems such as concrete, masonry, wood, and steel.
- Describe the fundamental concepts and benefits of BIM in the architecture, engineering, construction, and operations (AECO) industry.
- Distinguish BIM from traditional CAD-based design methods.
- Create basic architectural and structural components in Revit.
- Modify and organize model elements to support design intent and documentation.
- Apply basic annotation, detailing, and sheet layout functions in Revit.
- Integrate model views and schedules for project communication.

Course Structure and evaluation

This course is a sixteen-week course. Sections are provided in the learning management systems with assignments for each module. Each section's assignment must be completed by due date and time. After the due date, the instructor will evaluate all student submissions.

The evaluation is based on engagement in-class activities, homework assignments, quizzes and final project. This course uses an active learning approach, in which students should be active outside of the classroom for learning. During the class time of the BIM section of the course we will have hands-on experiences to focus information modeling using Autodesk Revit.

Grading Structure

Content	Percentage
Print Reading assignments and quizzes	25%
Print Reading groupworks	15%
Revit assignment	35%
Team Project	15%
Attendance and Participation	10%
Total	100%

Grading Scale:

Individual grade items and your final grade will be assessed as follows:

A = 100- 90%

B = 89- 80%

C = 79-70%

D = 69-60%

F = less than 60%

Grading Policies

Deductions to your assignments grade for late submission will be given as follows:

0-24 hours late a deduction of 15% of the earned grade

24-48 hours late a deduction of 40% of the earned grade

48-72 hours late a deduction of 60% of the earned grade

More than 72 hours late No credit.

Assignments must still be submitted to get a grade for this course.

Course outline:

* The course schedule is subject to change throughout the semester. Revisions will be noted in class.

Week	Date	Class topics	Readings / Assignments & Due Date
Week 1	Aug 26 Aug 28	Overview of Class Syllabus Introduction to Book and Plans Construction Drawing Organization Plan-set Information Types of Prints Uniform Drawing System	Units 1 & 4
Week 2	Sept 2 Sept 4	Plan Lines and Symbols Alphabet of Lines Construction Drawings Construction Math Measuring Drawing Scales	Units 2 - 5
Week 3	Sept 9 Sept 11	Reading Prints Site Plans Plan Views Architectural Drawings	Units 5 & 8
Week 4	Sept 16 Sept 18	Foundations & Structural Prints Footings, grade beams, slabs Structural components	Units 10 & 11
Week 5	Sept 23 Sept 25	Framing: Residential & Light Commercial Components of Framing Roof Framing Load-bearing Partitions	Unit 12
Week 6	Sept 30 Oct 2	MEP Mechanical Plans Electrical Drawings Plumbing Prints	Units 13, 14 & 15
Week 7	Oct 7 Oct 9	Specifications & Materials Specifications CSI Format Welding Symbols Estimating	Units 6, 7, 16 & 17
Week 8	Oct 14 Oct 16	Introduction and overview of BIM Differences between BIM and CAD How is BIM changing the AEC industry? Uses of BIM Introduction to Revit Navigation	
Week 9	Oct 21 Oct 23	BIM Authoring – Architectural modeling (Modeling basics, setting up projects, modeling basic building exterior objects)	
Week 10	Oct 28 Oct 30	BIM Authoring – Architectural modeling Object hierarchy Modeling Process for an Example Building	
Week 11	Nov 4 Nov 6	BIM Authoring – Architectural modeling Modeling Process for an Example Building	
Week 12	Nov 11 Nov 13	BIM Authoring – Family Creation Custom Modeling	
Week 13	Nov 18 Nov 20	BIM Authoring – Structural Modeling Workshop	
Week 14		Thanksgiving Break	
Week 15	Dec 2 Dec 4	Schedules and Quantities Workshop	

Week 16	Dec 8 Dec 10	Work on Final Project	
Final exam week		No Final Exam	

Note: This is a tentative schedule, and subject to change as necessary. The instructor reserves the right to make changes to the topics or sequence of topics, schedule, assignments, grading plan, and content of the course.

Academic Integrity:

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- AI expectations: Check the FDIC recommendation https://www.eiu.edu/fdic/ai_guidance.php

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Skill Requirements

An Internet connection is necessary to participate in the assignments, access readings, transfer course work, and receive feedback from your instructional associate and/or professor. For this course, students should have a basic working knowledge of computers and Internet use as well as access to a computer with a broadband (DSL, cable, satellite) Internet connection. At a minimum, you must have a computer with Windows 7.0 or MAC 10.6. Microsoft Office 2010 including Microsoft Word, PowerPoint, and Excel are necessary. Adobe Acrobat may be used only if necessary, and with the permission of the instructor. Microsoft Office is the standard office productivity software utilized by EIU faculty, students, and staff. Microsoft Word is the standard word processing software, Microsoft Excel is the standard spreadsheet software, and Microsoft PowerPoint is the standard presentation software. Copying and pasting, along with attaching/uploading documents for assignment submission, will also be required.



Course syllabus

TEC 5143- Research in Technology

Term: Fall 2025

Course Time: Tuesday: 6:00 PM to 8:30 PM

Classroom: Klehm Hall 4431

Course Dates: Aug. 25, 2025 – Dec. 12, 2025

Faculty: Dr. Reachsak Ly

Email: rlly@eiu.edu (Preferred communication method)

Office: Klehm Hall 4014

Office Hours: Tuesday and Thursday, 11:00 AM to 1:00 PM or by appointment

Communication Response Time: Within 2 days

Required Textbooks

Practical Research: Planning and Design, (11th Edition), Paul D. Leedy; Jeanne Ellis Ormrod Publisher ISBN: 9780133741322 ([Textbook](#))

Course description

This course introduces the means and methods of research used to define and investigate problems in technology-related fields. Topics include problem and scope definition, literature review, research methodologies, data collection, and data analysis. Students will learn the processes and tools for conducting both experimental and non-experimental research, with an emphasis on research design, analysis, interpretation, and reporting. The course also guides students in the development and writing of a research proposal.

Course objectives

Objectives of this course is to prepare students to conduct research in different technological fields. Students will be able to learn to define a research problem, to conduct effective research using library and online resources, write literature reviews and research reports, and to present their research findings in a professional manner in a group environment.

Student Learning outcome

At the end of the course the student should be able to understand the knowledge and skills required to:

1. Generate a research topic and a research problem statement and develop an empirical research question
2. Use the reference area of the library, interlibrary loan services, and computerized databases to locate relevant articles on a selected topic.
3. Execute literature reviews, Search and analyze literature to identify a gap in empirical knowledge.
4. Formulate a hypothesis statement.
5. Critical read reviews of research and critique as to:
 - a. Definition of a topic or problem
 - b. Adequacy of prior reviews in the area

- c. Selection of studies to be reviewed.
- d. Analysis of study findings and pertinent characteristics
- 6. Develop a system for collecting data about the characteristics and results of previously conducted research.
- 7. Analyze this collected data and interpret and report the results.
- 8. Use APA standards when writing a research proposal.
- 9. Develop a deliverable Research Portfolio that can be used in further research.
- 10. Describe ethical issues related to empirical research and expectations for ethical behavior
- 11. Describe the steps required to create an effective empirical research proposal
- 12. Evaluate appropriateness of data collection and analytical methods to answer empirical research questions

Grading Structure

Content	Percentage
Attendance and participation	10%
Literature review	15%
PowerPoint presentation	15%
Quizzes	10%
Journal entries	10%
Proposal related assignment	15%
Final Proposal	10%
In-class activities	15%
Total	100%

Grading Scale:

Individual grade items and your final grade will be assessed as follows:

A = 100- 90%

B = 89- 80%

C = 79-70%

D = 69-60%

F = less than 60%

Grading Policies

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24-48 hours late a deduction of 40% of the earned grade

48-72 hours late a deduction of 60% of the earned grade

More than 72 hours late No credit.

Assignments must still be submitted to get a grade for this course.

Reading Assignments

Students are expected to read the textbook chapters and course slides in the relevant Module before starting each course session on the D2L.

Participation

Face to Face (F2F) courses students will need to attend class and participate in classroom discussions or activities in order to receive participation points. We will have either have activities or online based activities to show participation for the week. Partial participation points will be given in certain approved circumstances with official proof provided, (i.e., doctors note, accident report, etc.)

Journal Entries

Journal entries are not for reciting facts. It is a self-reflection on what you have learned and how you plan to use or have used the information. You are required to complete a journal entry every week for 12 weeks during the semester. See the “Journal Entry Requirements” PDF document in the Assignment Explanation Module.

Literature Reviews

Each student is asked to provide a list of three scholarly articles that are relevant to his/her own research topic interest. Students will use these articles to create their qualitative, or quantitative research proposals. This exercise is also designed to provide the opportunity for students to demonstrate their skill at completing some basic steps necessary to conduct literature reviews. See the “Literature Review Requirements - Sample” PDF document in the Assignment Explanation Module.

PowerPoint Presentations

Each student will present their Research Project in a series of two PowerPoint presentations. The first will be a 5-minute presentation based on their Research Topic, the Research Problem, Sub-problems and your Hypothesis. This will be completed on Week 7. The second will be a 10-minute presentation based on their Proposal Introduction, Problem Statement, Objectives, Preliminary Literature Review, and Methodology. This will be completed in the final class meetings. Attendance is mandatory, even if you have already presented your Research.

Quizzes

The quiz date will be announced in advance and will cover material from specified chapters in the textbook and from PowerPoint slides in the course Modules. Each quiz will be worth 25 points based of 25 questions from True and False or Multiple Choice. Each quiz will be available on D2L.

Research Project Proposal

There will be an individual project proposal that each student will write based on the instructions provided in the D2L. There will be a series of assignments based on the components required to create a proposal. Students will prepare and submit their components in an approved format and APA style in a Word document. The details of the project will be announced on D2L throughout the semester. See the “A sample proposal with comment” and “Sample Proposal Format” PDF documents in the Assignment Explanation Module.

In-class Activities

Some in-class activities and additional assignments (Library, APA and Proposal Correction) are required during the semester. You are responsible to read the description of these assignments on D2L and complete them with instructions when the appropriate Module is discussed. Some of these exercises will require submission via d2l.

Course outline

* The course schedule is subject to change throughout the semester. Revisions will be noted in class.

Week	Date	Class topics	Reading	Assignments and Quizzes
Week 1	Aug 26	Course introduction Introduction to empirical research		

Week 2	Sept 2	The Nature and Tools of Research	Chapter 1	
Week 3	Sept 9	The Problem: The Heart of the Research Process Problem Statements	Chapter 2	
Week 4	Sept 16	Review of the Related Literature Literature Analysis Gap statement	Chapter 3	
Week 5	Sept 23	Planning your Research Project Research Questions	Chapter 4	
Week 6	Sept 30	Writing the Research Proposal	Chapter 5	
Week 7	Oct 7	Presentations Quiz #1		
Week 8	Oct 14	Quantitative Descriptive Research	Chapter 6	
Week 9	Oct 21	Experimental, Quasi-Experimental, and Ex Post Facto Designs	Chapter 7	
Week 10	Oct 28	Analyzing Quantitative Data	Chapter 8	
Week 11	Nov 4	Qualitative Research Methods Quiz #2	Chapter 9	
Week 12	Nov 11	Historical Research	Chapter 10	
Week 13	Nov 18	Analyzing Qualitative Data	Chapter 11	
Week 14		Thanksgiving Break		
Week 15	Dec 2	Mixed-Methods Designs Limitations	Chapter 12	
Week 16	Dec 8	Final Presentations		
Final exam week		No Final Exam		

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- **AI and Academic Integrity:** Students are not allowed to use advanced automated tools (artificial intelligence or machine learning tools such as ChatGPT, Grammarly, or Dall-E 2) on assignments in this

course. Each student is expected to complete each assignment without substantive assistance from others, including automated tools.

Written work may be submitted through Turnitin in D2L. Turnitin includes an AI detection tool which indicates the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was generated by AI. This information may be used, in addition to other evidence, if there are concerns with academic misconduct.

It is also the responsibility of the student to ensure that his or her work does not qualify as plagiarism. If a student does not understand plagiarism, then he or she must seek out information on the matter. The excuse of ignorance on the matter is not an acceptable excuse.

Students with Disabilities

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EASTERN ILLINOIS UNIVERSITY

Fall 2025 - School of Technology

Faculty/Course Evaluation

Reachsak Ly

CMG 1000-001 (#94756 FA25)

No. of responses = 11

No. of enrolled = 23

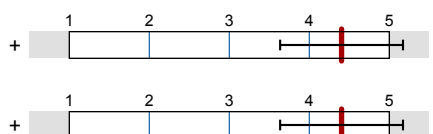
Rate of return = 47.83%



Overall indicators

Global Index

1. Evaluation ($\alpha = 0.99$)



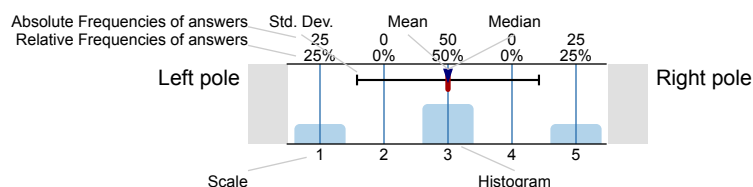
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av.=4.4
dev.=0.77

Survey Results

Legend

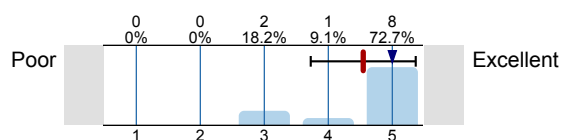
Question text



n=No. of responses
av.=Mean
md=Median
dev.=Std. Dev.
ab.=Abstention

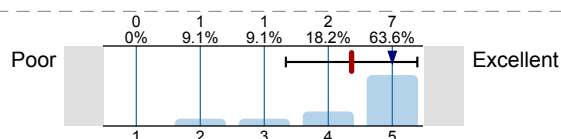
1. Evaluation

- 1.1) The instructor demonstrates command of the subject matter or discipline.



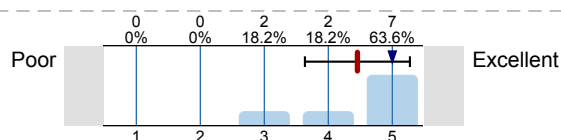
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md=5
dev.=0.82

- 1.2) The instructor effectively organizes knowledge or material for teaching and learning.



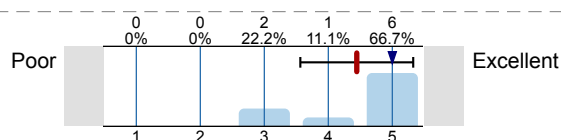
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dev.=1.03

- 1.3) The instructor presents knowledge or material effectively.



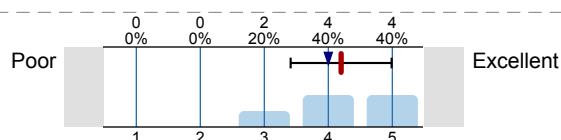
n=11
av.=4.45
md=5
dev.=0.82

- 1.4) The instructor explains lab procedures clearly.
(Do not respond to this item if this is not a lab course.)



n=9
av.=4.44
md=5
dev.=0.88

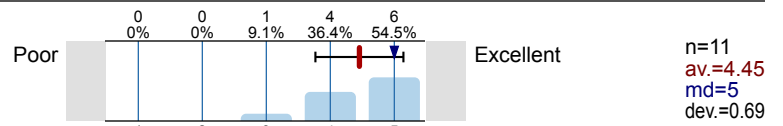
- 1.5) The instructor is in control of the direction of the class.



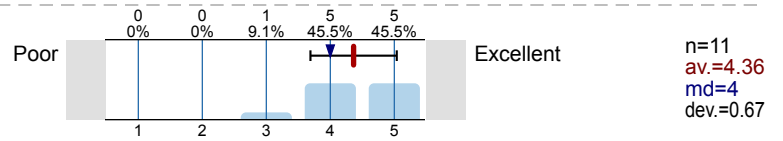
n=10
av.=4.2
md=4
dev.=0.79

1.6) The instructor appears to keep up to date on developments in the field.		n=11 av.=4.36 md=5 dev.=0.81
1.7) The instructor encourages and interests students in the learning process.		n=11 av.=4.27 md=5 dev.=0.9
1.8) The instructor encourages careful, critical, and independent thinking about the course materials.		n=11 av.=4.36 md=5 dev.=0.81
1.9) The instructor demonstrates proficient oral communication skills.		n=11 av.=4.27 md=5 dev.=0.9
1.10) The instructor's personal mannerisms are not annoying or distracting.		n=11 av.=4.36 md=4 dev.=0.67
1.11) The instructor is courteous and approachable.		n=11 av.=4.45 md=5 dev.=0.69
1.12) The instructor is readily accessible outside of class for face-to-face course sections or electronically for technology delivered course sections.		n=11 av.=4.36 md=4 dev.=0.67
1.13) The instructor respects differences of opinion.		n=11 av.=4.45 md=5 dev.=0.69
1.14) The instructor respects cultural diversity in the classroom.		n=11 av.=4.55 md=5 dev.=0.69
1.15) The instructor is actively helpful when students have problems.		n=11 av.=4.55 md=5 dev.=0.69
1.16) The course content is related to the course objective.		n=11 av.=4.45 md=5 dev.=0.69
1.17) The instructor establishes clear standards for grading.		n=11 av.=4.36 md=4 dev.=0.67

- 1.18) Exams and evaluation procedures are coordinated with what is taught and assigned in the course.



- 1.19) In general, this class is conducted in a manner conducive to learning.



Profile

Subunit: STEC F2F - Department

Name of the instructor: Reachsak Ly

Name of the course: CMG 1000-001
(Name of the survey)

Values used in the profile line: Mean

1. Evaluation

1.1) The instructor demonstrates command of the subject matter or discipline.	Poor					Excellent	n=11	av.=4.55
1.2) The instructor effectively organizes knowledge or material for teaching and learning.	Poor					Excellent	n=11	av.=4.36
1.3) The instructor presents knowledge or material effectively.	Poor					Excellent	n=11	av.=4.45
1.4) The instructor explains lab procedures clearly. (Do not respond to this item if this is not a lab course.)	Poor					Excellent	n=9	av.=4.44
1.5) The instructor is in control of the direction of the class.	Poor					Excellent	n=10	av.=4.2
1.6) The instructor appears to keep up to date on developments in the field.	Poor					Excellent	n=11	av.=4.36
1.7) The instructor encourages and interests students in the learning process.	Poor					Excellent	n=11	av.=4.27
1.8) The instructor encourages careful, critical, and independent thinking about the course materials.	Poor					Excellent	n=11	av.=4.36
1.9) The instructor demonstrates proficient oral communication skills.	Poor					Excellent	n=11	av.=4.27
1.10) The instructor's personal mannerisms are not annoying or distracting.	Poor					Excellent	n=11	av.=4.36
1.11) The instructor is courteous and approachable.	Poor					Excellent	n=11	av.=4.45
1.12) The instructor is readily accessible outside of class for face-to-face course sections or electronically for technology delivered course sections.	Poor					Excellent	n=11	av.=4.36
1.13) The instructor respects differences of opinion.	Poor					Excellent	n=11	av.=4.45
1.14) The instructor respects cultural diversity in the classroom.	Poor					Excellent	n=11	av.=4.55
1.15) The instructor is actively helpful when students have problems.	Poor					Excellent	n=11	av.=4.55
1.16) The course content is related to the course objective.	Poor					Excellent	n=11	av.=4.45
1.17) The instructor establishes clear standards for grading.	Poor					Excellent	n=11	av.=4.36
1.18) Exams and evaluation procedures are coordinated with what is taught and assigned in the course.	Poor					Excellent	n=11	av.=4.45
1.19) In general, this class is conducted in a manner conducive to learning.	Poor					Excellent	n=11	av.=4.36

Profile Line for Indicators

Subunit: STEC F2F - Department

■ Name of the instructor: Reachsak Ly

■ Name of the course:
(Name of the survey) CMG 1000-001

1. Evaluation



av.=4.4

dev.=0.77

Comments Report

2. Comments

2.1) What activities/topics were most useful and should be retained?

- all
- All of the topics
- I thought it was most useful whenever Dr. Ly would explain a activity in the industry like bidding and estimating. He would then show slides of a project and how to construct a bid on this specific project. We would then be given a practice worksheet. He is very helpful.
- Learning about all things you need to know in this field of work.
- project management
- The most useful topics were project planning, scheduling, cost control and practical activities such as case studies, because they helped connect theory with real construction management practices. These topics and activities should be retained in the course.
- When he broke down each and every activity. Gave everyone work time during class/ in class assignments

2.2) What activities/topics seemed to have the least relevance?

- Everything was useful
- Honestly, none. They all was very useful.
- n/a
- no comment
- None (2 Counts)
- Some of the more theoretical topics seemed to have the least relevance, especially those that were not directly connected to real world construction projects or practical applications.

2.3) What other comments do you want to make about the teaching of this course?

- An excellent teacher would have him again.
- He's a very good professor
- N/A
- no comment
- None
- The course was well organized and the instructor explained the material clearly. Including more real world examples and hands on activities would further improve the learning experience.

EASTERN ILLINOIS UNIVERSITY

Fall 2025 - School of Technology

Faculty/Course Evaluation

Reachsak Ly

CMG 2223-001 (#97285 FA25)

No. of responses = 11

No. of enrolled = 23

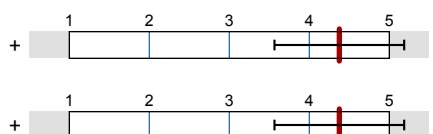
Rate of return = 47.83%



Overall indicators

Global Index

1. Evaluation ($\alpha = 0.98$)



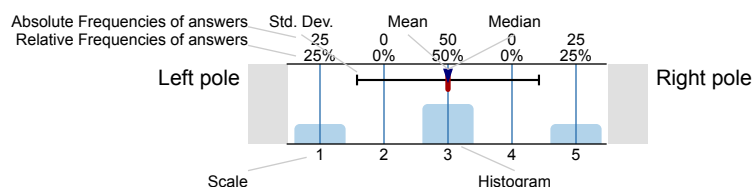
av.=4.38
dev.=0.81

av.=4.38
dev.=0.81

Survey Results

Legend

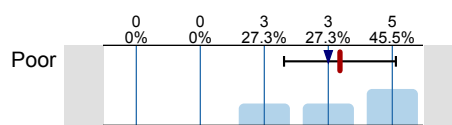
Question text



n=No. of responses
av.=Mean
md=Median
dev.=Std. Dev.
ab.=Abstention

1. Evaluation

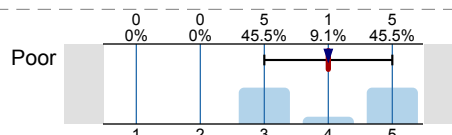
- 1.1) The instructor demonstrates command of the subject matter or discipline.



Excellent

n=11
av.=4.18
md=4
dev.=0.87

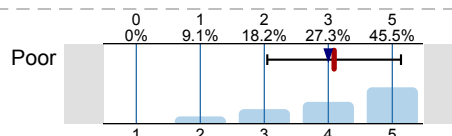
- 1.2) The instructor effectively organizes knowledge or material for teaching and learning.



Excellent

n=11
av.=4
md=4
dev.=1

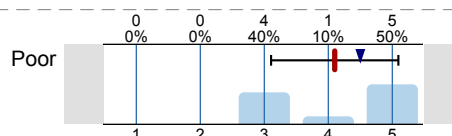
- 1.3) The instructor presents knowledge or material effectively.



Excellent

n=11
av.=4.09
md=4
dev.=1.04

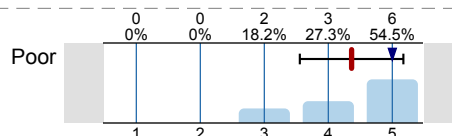
- 1.4) The instructor explains lab procedures clearly.
(Do not respond to this item if this is not a lab course.)



Excellent

n=10
av.=4.1
md=4.5
dev.=0.99

- 1.5) The instructor is in control of the direction of the class.

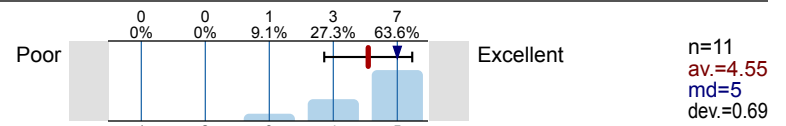


Excellent

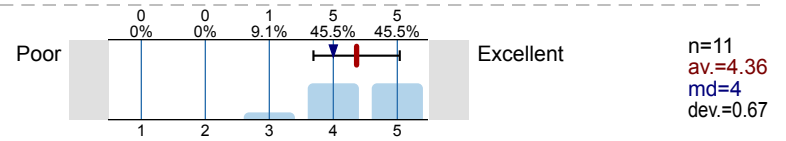
n=11
av.=4.36
md=5
dev.=0.81

1.6) The instructor appears to keep up to date on developments in the field.		n=11 av.=4.45 md=5 dev.=0.82
1.7) The instructor encourages and interests students in the learning process.		n=11 av.=4.36 md=5 dev.=0.81
1.8) The instructor encourages careful, critical, and independent thinking about the course materials.		n=11 av.=4.27 md=4 dev.=0.79
1.9) The instructor demonstrates proficient oral communication skills.		n=11 av.=3.82 md=4 dev.=1.25
1.10) The instructor's personal mannerisms are not annoying or distracting.		n=11 av.=4.55 md=5 dev.=0.69
1.11) The instructor is courteous and approachable.		n=11 av.=4.64 md=5 dev.=0.67
1.12) The instructor is readily accessible outside of class for face-to-face course sections or electronically for technology delivered course sections.		n=11 av.=4.64 md=5 dev.=0.67
1.13) The instructor respects differences of opinion.		n=11 av.=4.64 md=5 dev.=0.67
1.14) The instructor respects cultural diversity in the classroom.		n=11 av.=4.64 md=5 dev.=0.67
1.15) The instructor is actively helpful when students have problems.		n=11 av.=4.55 md=5 dev.=0.69
1.16) The course content is related to the course objective.		n=11 av.=4.64 md=5 dev.=0.67
1.17) The instructor establishes clear standards for grading.		n=11 av.=4.27 md=5 dev.=0.9

- 1.18) Exams and evaluation procedures are coordinated with what is taught and assigned in the course.



- 1.19) In general, this class is conducted in a manner conducive to learning.



Profile

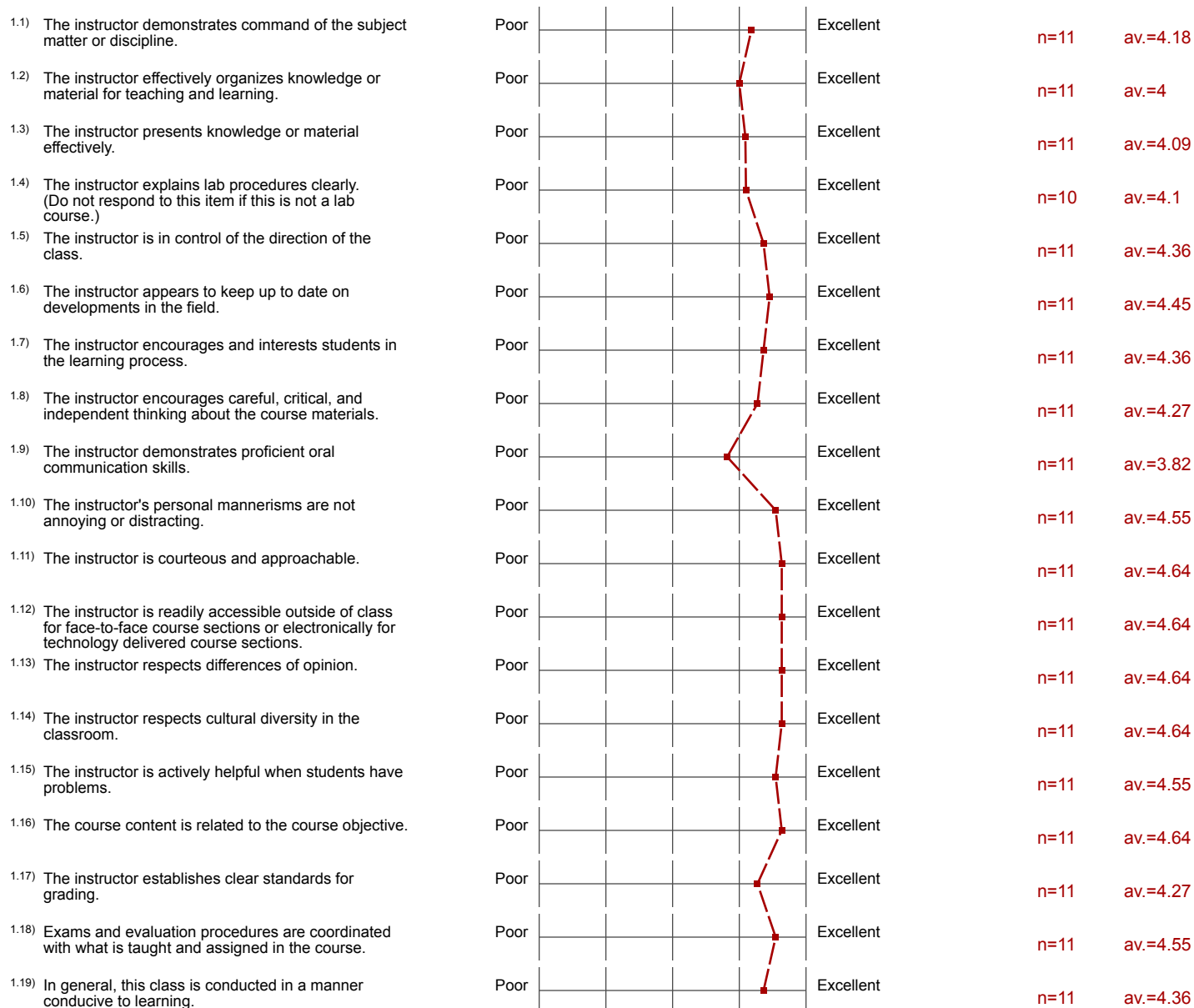
Subunit: STEC F2F - Department

Name of the instructor: Reachsak Ly

Name of the course: CMG 2223-001
(Name of the survey)

Values used in the profile line: Mean

1. Evaluation



Profile Line for Indicators

Subunit: STEC F2F - Department

■ Name of the instructor: Reachsak Ly

■ Name of the course:
(Name of the survey) CMG 2223-001

1. Evaluation



av.=4.38

dev.=0.81

Comments Report

2. Comments

2.1) What activities/topics were most useful and should be retained?

- all good
- BIM
- everything in this course is vital to the development of future construction management majors.
- learning about blueprinting then actually making the plans ourselves.
- Learning how to use revit and being able to build structures. This could be very useful in a job.
- The activities that were most useful and should definitely be retained were the in-class assignments combined with open question-and-answer time. Working through assignments together during class helped reinforce the material immediately, instead of leaving all of the learning for outside homework. Being able to ask questions in real time when something was confusing made the topics much easier to understand and prevented small misunderstandings from turning into bigger problems later on.

In addition, the worksheets that were given after or alongside lessons were very helpful for further increasing understanding. They allowed us to practice the concepts on our own while still being closely tied to what we had just learned in class. Keeping this balance between guided, in-class work and independent worksheets was especially effective and helped solidify the material.

- The Revit section is a great introduction to the software and should be retained.
- the walkthrough videos on d2l
- Using revit

2.2) What activities/topics seemed to have the least relevance?

- all good
- every topic had its respective relevance
- none (2 Counts)
- None (2 Counts)
- Projects
- some of the introductory blueprint readings
- The print reading section is useful but it is introduced with large plans that make it difficult for a student to really understand the topic.

2.3) What other comments do you want to make about the teaching of this course?

- all good
- He's a very good teacher and is funny. makes learning fun.
- I wish the class focused more on hands reading plans/drawing up plans. Drawing up the plans has been very interesting and new to me. Having less PowerPoint sessions on reading the plans would have been nice. I'm not sure what a good alternative would be, but walking through the reading plans section more thoroughly would've been nice.
- None (3 Counts)
- none
- Overall the course is very useful and the professor did a decent job of expanding construction knowledge in Print Reading and BIM.
- very good professor for being so young, and super helpful

EASTERN ILLINOIS UNIVERSITY

Fall 2025 - School of Technology
Faculty/Course Evaluation

Reachsak Ly

TEC 5143-003 (#98071 FA25)

No. of responses = 4

No. of enrolled = 6

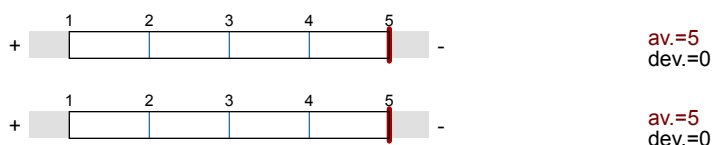
Rate of return = 66.67%



Overall indicators

Global Index

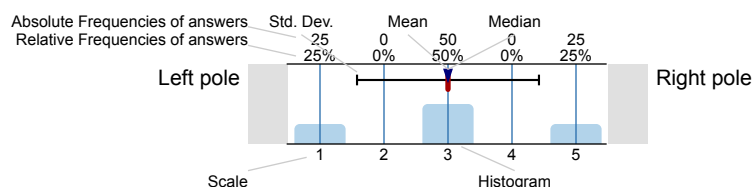
1. Evaluation



Survey Results

Legend

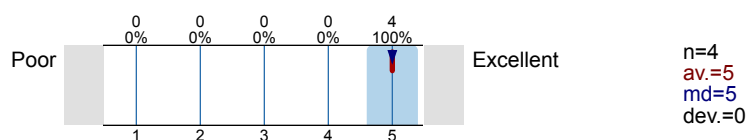
Question text



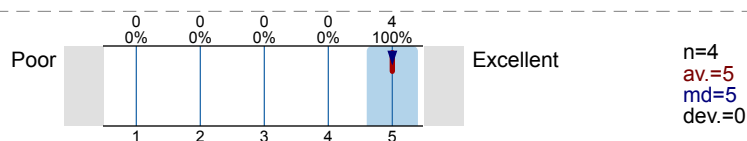
n=No. of responses
av.=Mean
md=Median
dev.=Std. Dev.
ab.=Abstention

1. Evaluation

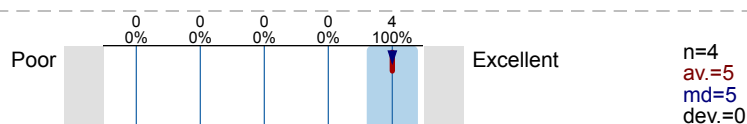
- 1.1) The instructor demonstrates command of the subject matter or discipline.



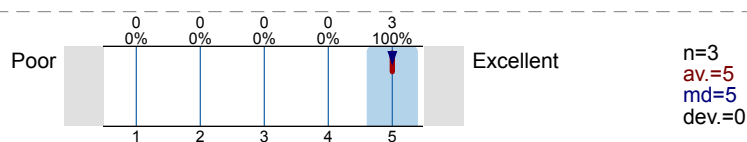
- 1.2) The instructor effectively organizes knowledge or material for teaching and learning.



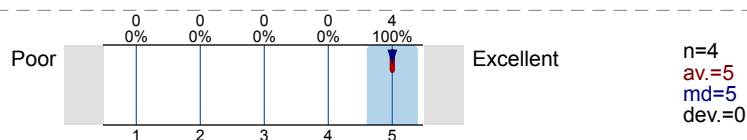
- 1.3) The instructor presents knowledge or material effectively.



- 1.4) The instructor explains lab procedures clearly.
(Do not respond to this item if this is not a lab course.)

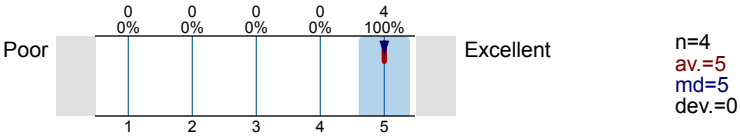


- 1.5) The instructor is in control of the direction of the class.

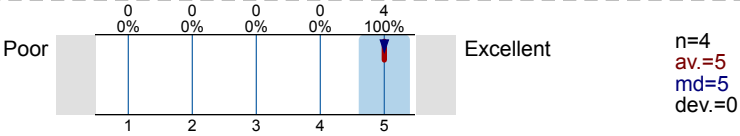


1.6) The instructor appears to keep up to date on developments in the field.		n=4 av.=5 md=5 dev.=0
1.7) The instructor encourages and interests students in the learning process.		n=4 av.=5 md=5 dev.=0
1.8) The instructor encourages careful, critical, and independent thinking about the course materials.		n=4 av.=5 md=5 dev.=0
1.9) The instructor demonstrates proficient oral communication skills.		n=4 av.=5 md=5 dev.=0
1.10) The instructor's personal mannerisms are not annoying or distracting.		n=4 av.=5 md=5 dev.=0
1.11) The instructor is courteous and approachable.		n=4 av.=5 md=5 dev.=0
1.12) The instructor is readily accessible outside of class for face-to-face course sections or electronically for technology delivered course sections.		n=4 av.=5 md=5 dev.=0
1.13) The instructor respects differences of opinion.		n=4 av.=5 md=5 dev.=0
1.14) The instructor respects cultural diversity in the classroom.		n=4 av.=5 md=5 dev.=0
1.15) The instructor is actively helpful when students have problems.		n=4 av.=5 md=5 dev.=0
1.16) The course content is related to the course objective.		n=4 av.=5 md=5 dev.=0
1.17) The instructor establishes clear standards for grading.		n=4 av.=5 md=5 dev.=0

1.18) Exams and evaluation procedures are coordinated with what is taught and assigned in the course.



1.19) In general, this class is conducted in a manner conducive to learning.



Profile

Subunit: STEC F2F - Department

Name of the instructor: Reachsak Ly

Name of the course: TEC 5143-003
(Name of the survey)

Values used in the profile line: Mean

1. Evaluation

1.1) The instructor demonstrates command of the subject matter or discipline.	Poor					Excellent	n=4	av.=5
1.2) The instructor effectively organizes knowledge or material for teaching and learning.	Poor					Excellent	n=4	av.=5
1.3) The instructor presents knowledge or material effectively.	Poor					Excellent	n=4	av.=5
1.4) The instructor explains lab procedures clearly. (Do not respond to this item if this is not a lab course.)	Poor					Excellent	n=3	av.=5
1.5) The instructor is in control of the direction of the class.	Poor					Excellent	n=4	av.=5
1.6) The instructor appears to keep up to date on developments in the field.	Poor					Excellent	n=4	av.=5
1.7) The instructor encourages and interests students in the learning process.	Poor					Excellent	n=4	av.=5
1.8) The instructor encourages careful, critical, and independent thinking about the course materials.	Poor					Excellent	n=4	av.=5
1.9) The instructor demonstrates proficient oral communication skills.	Poor					Excellent	n=4	av.=5
1.10) The instructor's personal mannerisms are not annoying or distracting.	Poor					Excellent	n=4	av.=5
1.11) The instructor is courteous and approachable.	Poor					Excellent	n=4	av.=5
1.12) The instructor is readily accessible outside of class for face-to-face course sections or electronically for technology delivered course sections.	Poor					Excellent	n=4	av.=5
1.13) The instructor respects differences of opinion.	Poor					Excellent	n=4	av.=5
1.14) The instructor respects cultural diversity in the classroom.	Poor					Excellent	n=4	av.=5
1.15) The instructor is actively helpful when students have problems.	Poor					Excellent	n=4	av.=5
1.16) The course content is related to the course objective.	Poor					Excellent	n=4	av.=5
1.17) The instructor establishes clear standards for grading.	Poor					Excellent	n=4	av.=5
1.18) Exams and evaluation procedures are coordinated with what is taught and assigned in the course.	Poor					Excellent	n=4	av.=5
1.19) In general, this class is conducted in a manner conducive to learning.	Poor					Excellent	n=4	av.=5

Profile Line for Indicators

Subunit:

STEC F2F - Department

Name of the instructor:

Reachsak Ly

Name of the course:

TEC 5143-003

(Name of the survey)

1. Evaluation



av.=5

dev.=0

Comments Report

2. Comments

^{2.1)} What activities/topics were most useful and should be retained?

- The most useful activities in this course were the research-related assignments, including the Research Topic, Research Problem, Initial Martini Glass Framework, Literature Reviews, and the Proposal sections. These activities helped me build my research project step-by-step and made the entire process easier to understand. The in-class activities, such as the literature search and statistical analysis, were also very helpful because they provided hands-on practice. I believe these assignments and activities are essential to student learning and should definitely be retained.
- The most useful parts of the course were the guidance on building a research framework, forming problem statements, and understanding methodologies. The step-by-step breakdown of each research element, like literature reviews, hypotheses, and data collection, was especially helpful and should definitely be retained.
- The topics on various research methods are very much useful and needs to be retained

2.2) What activities/topics seemed to have the least relevance?

- None
- Overall, the course content was highly relevant and aligned with the learning objectives. If anything, a few of the shorter journal entries felt somewhat repetitive compared to the major research assignments. However, even these activities contributed to reinforcing understanding and developing skills, so nothing in the course felt truly irrelevant.
- There weren't any topics that felt irrelevant. All the activities connected well to the research process and supported the development of the final proposal.

2.3) What other comments do you want to make about the teaching of this course?

- The instructor was exceptional in guiding students through every stage of the research process. His explanations were clear, feedback was timely and constructive, and he consistently created a supportive and respectful learning environment. He encouraged critical thinking and independent analysis, making a challenging course both manageable and engaging. His dedication and approachability greatly enhanced the overall learning experience.
- The teaching was clear, structured, and supportive throughout the semester. The explanations made each part of the research process easier to understand, and the feedback was helpful for improving my work. Overall, the course created a strong foundation for conducting academic research.
- This course is very much useful and expands ones point of view about the ways of research and broadens the thinking process.



GREEN BUSINESS CERTIFICATION INC. CERTIFIES THAT

REACHSAK LY

HAS ATTAINED THE DESIGNATION OF

LEED[®] Green Associate[™]

by demonstrating the knowledge and understanding of
green building practices and principles needed to
support the use of the LEED green building program.

11679119-GREEN-ASSOCIATE

CREDENTIAL ID

04 AUG 2025

ISSUED

03 AUG 2027

VALID THROUGH

A handwritten signature in black ink, reading "Peter Templeton".

PETER TEMPLETON
PRESIDENT & CEO
U.S. GREEN BUILDING COUNCIL & GREEN BUSINESS CERTIFICATION INC.



Office of Research and Innovation
3046 Torgersen (0217)
620 Drillfield Drive
Blacksburg, Virginia 24061
P: (540) 231-9171
F: (540) 231-3031
<https://eng.vt.edu/research.html>

February 28, 2025

Reachsak Ly - MLSoC

Dear Reachsak,

Congratulations! You have been awarded a fellowship from the College of Engineering Office of Research and Innovation in the amount of \$2,289.20. This fellowship is a testament to your outstanding research contributions and the high regard in which you are held by faculty in your department.

This fellowship provides a stipend of \$457.84 per pay period, starting 2/25/2025 and ending 5/9/2025. This award does not include payment of tuition or any fees.

Fellowships are not considered employment, and you will not receive a W2 form at the end of the year. Fellows are required to report the fellowship stipend income to IRS on a 1040 form. For more information see IRS Publication 970: *Tax Benefits for Education*.

Sincerely,

A handwritten signature in black ink, appearing to read 'Nicole Akers'.

Nicole Akers

Assistant Director of Research
Office of Research & Innovation
VT College of Engineering

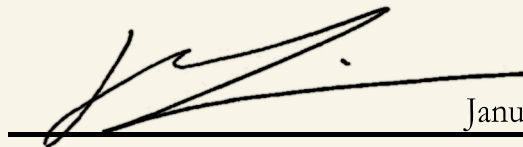
CERTIFICATE

This certificate is presented to

Jiawei Zhang, Jun Li, Reachsak Ly, Yunyi Liu, Jiangpeng Shu

*for winning the Excellence Award in the 1st International Project Competition for Structural Health Monitoring (IPC-SHM 2020) with paper entitled of
“Deep Learning-Based Fatigue Cracks Detection in Bridge Girders using Feature Pyramid Networks”*

Congratulations!



January 15, 2021

Prof. Hui Li, Chair
Harbin Institute of Technology

ANCRISST
Lab of Intelligent Civil Infrastructure, Harbin Institute of Technology, China
Smart Structures Technology Lab, University of Illinois at Urbana-Champaign, USA
CCCC Highway Consultants Co., Ltd.



January 15, 2021

Prof. Billie F. Spencer Jr, Co-Chair
University of Illinois at Urbana-Champaign

CCCC Civil Big Data Information Technology (Beijing) Co., Ltd.
State Key Lab for Health and Safety of Bridge Structures, China Railway Bridge Science Research Inst., Ltd.
State Key Lab on Safety and Health of In-service Long-span Bridges, JSTI Group, China

CERTIFICATE

OF APPRECIATION IS

Proudly Presented To

Reachsak Ly

in recognition of your exceptional dedication and exemplary service as a
Graduate Teaching Assistant during the 2024-2025 academic year.
With profound appreciation, we celebrate your unwavering commitment to
academic excellence, student success, and to the advancement of learning.



COLLEGE OF ENGINEERING
MYERS-LAWSON
SCHOOL OF CONSTRUCTION
VIRGINIA TECH.

CERTIFICATE OF SCHOLARLY RECOGNITION

Proudly Presented To

Reachsak Ly

for his outstanding academic achievement and dedication to scholarly excellence during the 2024-2025 academic year. Your work stands as a testament to the pivotal role of diligent inquiry and research in advancing our industry. We commend your commitment to the highest standards of academic pursuits and celebrate the impact of your exceptional achievements.



COLLEGE OF ENGINEERING
MYERS-LAWSON
SCHOOL OF CONSTRUCTION
VIRGINIA TECH.

Virginia Polytechnic Institute and State University

*The Board of Visitors of the Virginia Polytechnic Institute
and State University has conferred upon*

Reachsak Ly

upon the recommendation of the Faculty, the degree of

Graduate Certificate

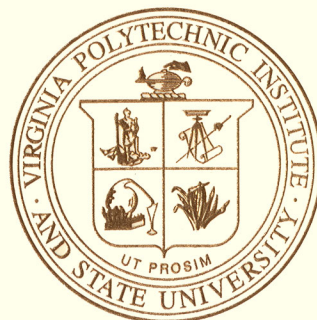
Future Professoriate

with all the rights, privileges and honors pertaining thereto.

*In testimony whereof, the undersigned, by authority
vested in them, have hereunto affixed their signatures
and the seal of the University at Blacksburg, Virginia
this fourteenth day of December, two thousand and twenty-three.*

Edward H. Baine

Rector

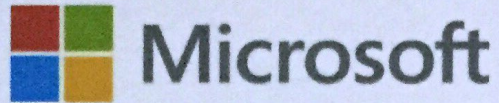


Timothy D. Sand

President

Aimee Juperant

Dean



Certificate of Participation

is awarded to

Rezeckir Ly

for participating in

“Visit Cambodia”

Windows App Studio Challenge for Destination Marketing

held on

10th March 2016

At Science & Engineering Festival Cambodia

A handwritten signature in black ink, appearing to read "David Lim".

David Lim

Director, Developer Evangelism & Experience, Microsoft APAC AHQ



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RESEARCH-ARTICLE

Smart Building Operations and Virtual Assistants Using LLM

REACHSAK LY, Virginia Tech College of Engineering, Blacksburg, VA, United States

ALIREZA SHOJAEI, Virginia Tech College of Engineering, Blacksburg, VA, United States

XINGHUA GAO, Virginia Tech College of Engineering, Blacksburg, VA, United States

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Reachsak Ly
Myers-Lawson School of
Construction,
Virginia Tech
Blacksburg, Virginia, USA
reachsak@vt.edu

Alireza Shojaei
Myers-Lawson School of
Construction,
Virginia Tech
Blacksburg, Virginia, USA
shojaei@vt.edu

Xinghua Gao
Myers-Lawson School of
Construction,
Virginia Tech
Blacksburg, Virginia, USA
xinghua@vt.edu

ABSTRACT

Conventional AI-powered smart home assistants primarily function as voice-activated control systems with limited adaptability and contextual understanding. Similarly, while traditional artificial intelligence has advanced autonomous building research, it often relies on predefined rules and struggles with real-time decision-making in dynamic building environments. This paper introduces a novel Generative AI-driven framework that integrates Large Language Models (LLMs) to create a smart generative AI-based virtual assistant and an operation automation system for building infrastructure. The AI systems autonomously manage building operations by analyzing real-time occupancy patterns and adjusting environmental conditions based on predefined comfort thresholds. The proposed system also facilitates seamless human-building interaction through an LLM-powered virtual assistant. The framework is validated through a prototype implementation in a real-world building equipped with smart appliances, with evaluations focusing on the AI systems' accuracy, reliability, and scalability. The findings demonstrate that the prototype system can autonomously adjust building conditions, optimize energy usage, and provide intelligent assistance for building operation tasks.

CCS CONCEPTS

• **Human-centered computing** → **Human-computer interaction (HCI)**

KEYWORDS

Generative AI, Large Language Models, Smart Buildings, Autonomous Building Operations, Virtual Assistant

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1 Introduction

Research on autonomous buildings has become a promising frontier in the field of smart and sustainable infrastructure. Autonomous buildings are characterized by their ability to operate independently through self-management, self-sufficiency, and intelligent operation [1], [2]. Machine learning techniques and AI have played a pivotal role in enabling these advancements. Their capacity to analyze large volumes of data, recognize trends, and make well-informed choices has contributed significantly to improving energy efficiency, enhancing occupant comfort, and optimizing building performance [3], [4]. However, the functionality of this conventional machine learning model may rely on predefined rules or specific training data, which, in some circumstances, may not adequately capture the dynamic and complex nature of the building environment [5], [6]. Furthermore, they often struggle to adapt to evolving circumstances or to incorporate contextual information effectively [7].

The advent of large language models (LLMs) offers a promising avenue to overcome these limitations and unlock a new realm of possibilities for autonomous building operations. LLMs exhibit remarkable capabilities in natural language processing, which could enable seamless human-machine interactions and intelligent decision-making processes [8]. Unlike conventional machine learning algorithms such as deep learning, regression, clustering, and reinforcement learning, LLMs possess a deep understanding of contextual information and can engage in humanlike conversations [5], [6], which could allow for more intuitive and adaptive control of building systems. Despite its transformative potential, the application of LLMs in building automation and smart virtual assistants remains largely unexplored. This study aims to address these gaps by proposing a novel Generative AI-driven framework that seamlessly integrates LLMs into building automation systems for autonomous building management while providing an intelligent virtual assistant interface for enhanced human-building interaction.

2 Related work

Recent research has focused on applying machine learning to enhance smart home capabilities. For instance, researchers have proposed home automation based on activity recognition by using deep learning algorithms to recognize users' activities from accelerometer data [9]. Voice-based assistants have also been an

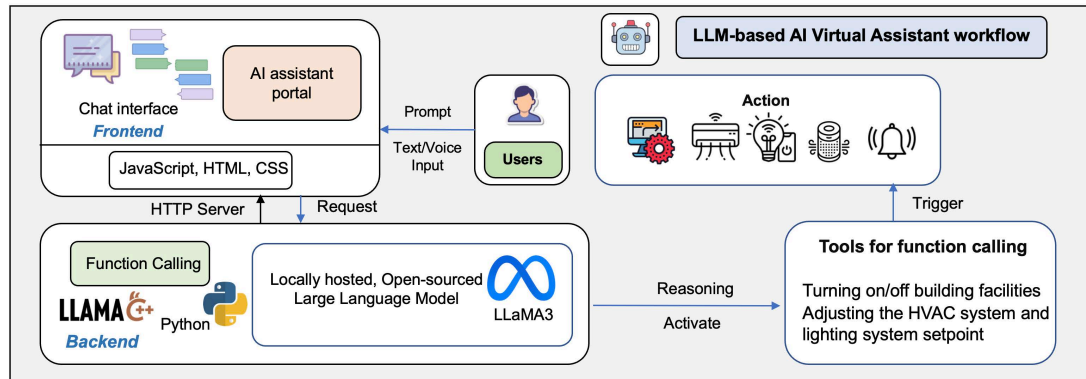


Figure 1: Overview of the LLM-based Virtual Assistant.

important focus area, with systems designed to understand users' voice commands through natural language processing (NLP). Rani et al. [10] developed a voice-controlled home automation system using NLP and IoT to control basic appliances. In addition, while commercial solutions like Google Assistant, Alexa, and Bixby offer user-friendly interfaces for handling various commands with advanced NLP techniques, they typically struggle with implicit and complex instructions [11]. These systems often fail when user utterances cannot be easily mapped to preprogrammed routines [12]. Web-based services like IFTTT [13], Zapier [14], and Home Assistant [15] enable users to create rules for controlling smart devices without extensive programming knowledge, though they lack the reasoning and context-awareness offered by LLMs.

To address these limitations, recent work has leveraged Large Language Models (LLMs) to better understand and execute complex user commands. Sasha [16] demonstrated that LLMs can produce reasonable behaviors in response to complex or vague commands by implementing a decision-making pipeline using LLM. However, Sasha's pipeline stages are manually defined and fixed, thereby limiting flexibility. Rivkin et al. [12] have also presented Smart Home Agent with Grounded Execution (SAGE), which overcomes these limitations using a scheme where user requests trigger an LLM-controlled sequence of discrete actions. In addition, LLM-powered autonomous agents are designed to perform complex tasks with minimal human intervention. Various agent architectures have been proposed [17], including Chain-of-Thought (CoT) [18], a prompting technique enabling complex reasoning through step-by-step planning while tools such as llama cpp agent [19] further extend LLM-based agent capabilities.

Furthermore, it is important to note that the existing research on the application of LLMs in the construction domain has primarily utilized commercialized GPT models, such as OpenAI's ChatGPT. The use of these cloud-based GPT models requires sensitive data to be transmitted to external servers for processing, potentially exposing it to unauthorized access or data breaches [20]. Another significant challenge lies in the cost and scalability aspects of these commercial LLM services. Therefore, there is also a need to explore alternative solutions, such as utilizing local and open-source LLMs that can improve inference speed while addressing data privacy concerns, as the data remains localized within the device or system. Our work builds upon these foundations while

also addressing limitations in flexibility, scalability, and integration with building infrastructure's automation system using open-sourced LLMs.

3 Methodology

This research employs the Design Science Research (DSR) [21] methodology to develop and validate the LLM-powered building automation system and virtual assistant. Previous studies [12], [22] reveal significant gaps in current building automation systems, including limited contextual understanding of traditional AI approaches and insufficient research on LLM integration in smart building applications. To address these gaps, the research objectives focus on developing an innovative framework that integrates LLMs into building automation systems for enhanced human-building interaction and autonomous environmental control. The design and development phase encompasses creating the LLM-powered virtual assistant interface, developing the autonomous building control system, and implementing real-time monitoring capabilities. The system will be demonstrated in a real-world smart building environment equipped with various sensors and smart appliances. Evaluation will assess the system's performance through multiple metrics, including environmental control accuracy, system reliability, and scalability. Finally, the research findings and framework design will be disseminated through academic publications.

4 Proposed framework

The proposed AI system consists of two primary components: the LLM-based Virtual assistant and the LLM-based AI agent. The Virtual assistant functions as a conversational interface that leverages LLM's language understanding capabilities for human-building interaction while the AI agent operates independently to monitor environmental conditions, analyze sensor data, and automate building operations without direct human input.

4.1 LLM-based Virtual Assistant

The LLM-based virtual assistant aims to facilitate the human-building interaction aspect within the proposed framework. Users can communicate with the virtual assistant through text and voice input to control various building facilities, adjust set points for the

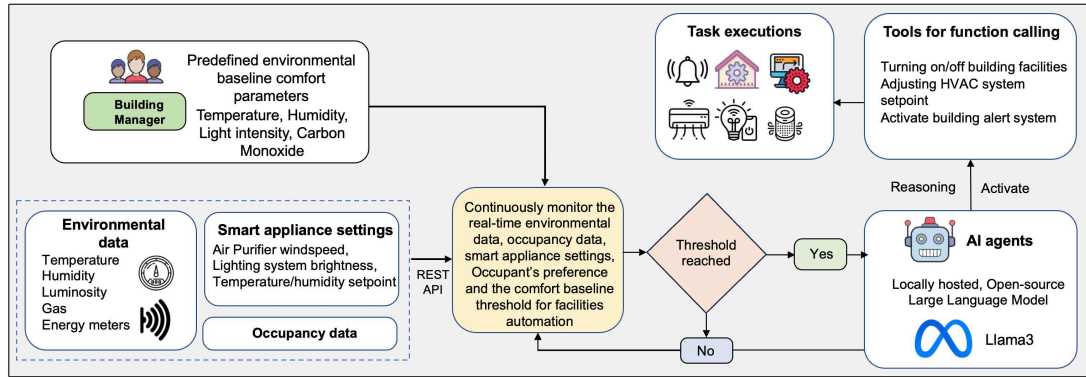


Figure 2: Overview of the LLM-powered autonomous building operation.

specific building smart facilities, or turn systems on or off as needed. Central to the virtual assistant is a locally hosted, open-source LLM and its function calling capabilities, specifically the LLaMA3 model by Meta [23]. The use of local and open-source LLM is driven by several factors, including enhanced data privacy and reduced operational costs. By keeping all interactions and data processing within the local infrastructure, the system also ensures independence from third-party entities and maintains strict control over sensitive information. The virtual assistant workflow is illustrated in Figure 1, which demonstrates the integration of frontend and backend components. Users interact with the system through a web-based chat interface. The submitted prompts by the user are transferred to the backend and processed by the local LLM model. Upon receiving a query, the LLM model leverages its reasoning and function-calling capabilities to understand the user's request and activates the appropriate predefined tools for task execution.

4.2 Autonomous building operation

The proposed LLM-based autonomous building operation framework leverages the integration of IoT devices and sensors, smart building facilities such as the HVAC and lighting system, and LLMs to create automated control of building systems (Figure 2). An array of environmental sensors and IoT Devices will be used to collect environmental data such as temperature, humidity, light intensity levels, carbon monoxide levels, as well as energy usage information, and occupancy levels within the physical space. Raspberry Pi devices are used to process these data and feed them into the AI agent through REST API. In addition, the threshold parameters are used for the automated smart building operation (e.g., max or min temperature, humidity, etc.). These threshold values are the baseline comfort parameters for ensuring optimal comfort and indoor environmental quality and are defined by users or building managers, who serve as administrators for the physical spaces. During the operation, the LLM-based AI agent will continuously compare real-time sensor data from the physical space against these predefined thresholds. When the environmental data values exceed these thresholds, the AI agent utilizes its function-calling capabilities to control various building systems, including smart lighting and HVAC systems. These operations involve

adjusting set points, activating or deactivating devices, or triggering alerts to maintain optimal building conditions.

5 Proof of Concept

5.1 Implementation preparation

5.1.1 Smart home appliance. To simulate the required environmental management capabilities for the study, a range of smart home devices has been installed, allowing the virtual assistant to control air quality, humidity, lighting, and temperature within the physical space. To simulate air quality control within the space, this study uses a smart air purifier device, Xiaomi Smart Air Purifier 4 Compact, which is equipped with multiple fan speed configurations. In addition, a smart humidifier, the Govee Smart Humidifier H7141, is used to facilitate the adjustment of humidity levels within the room. It offers multiple fan speed settings. The Xiaomi Mi Smart Standing Fan 2 is also integrated into the system by offering multiple fan speeds for personalized airflow control. This device allows the AI system to demonstrate its ability to regulate air circulation within the space, simulating traditional HVAC systems. For lighting control, this study used a smart bulb, Yeelight Smart Light Bulbs W3. It offers adjustable brightness levels, which mimic the control of indoor lighting conditions.

5.1.2 Deployment of the AI systems module. The proposed AI systems in this study are powered by LLaMA 3 8b, an open-source LLM with 8 billion parameters. The workstation used for the LLaMA3 model deployment in this study is an Apple MacBook Pro with an M1 Max chip and 32GB of RAM. To run the LLaMA 3 model efficiently, this study employs a quantized version of the model using the llama.cpp [24]. Llama.cpp is a tool that allows the execution of quantized LLMs on local hardware with support for different types of GPU. Quantization is an essential operation for the reduction of the model's size and computational requirements, which is particularly important for deploying AI systems on local hardware [25]. This enables running complex models on devices with less computational power without severely compromising performance. The 8-bit quantized model of LLaMA 3 8b is used in this study. Function calling is one of the core features of the proposed virtual assistant. In this context, function calling refers to the AI's ability to analyze user requests, extract key information,

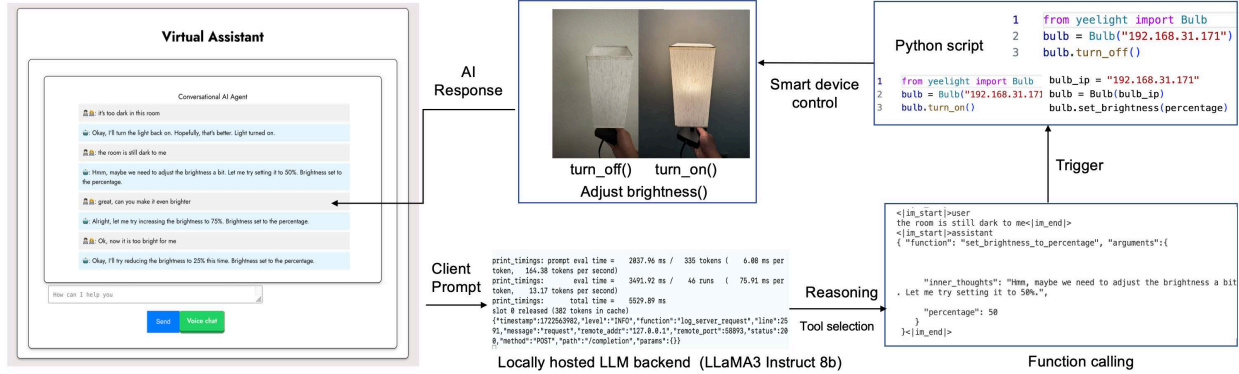


Figure 3: Smart building appliance control using LLM-based Virtual Assistant.

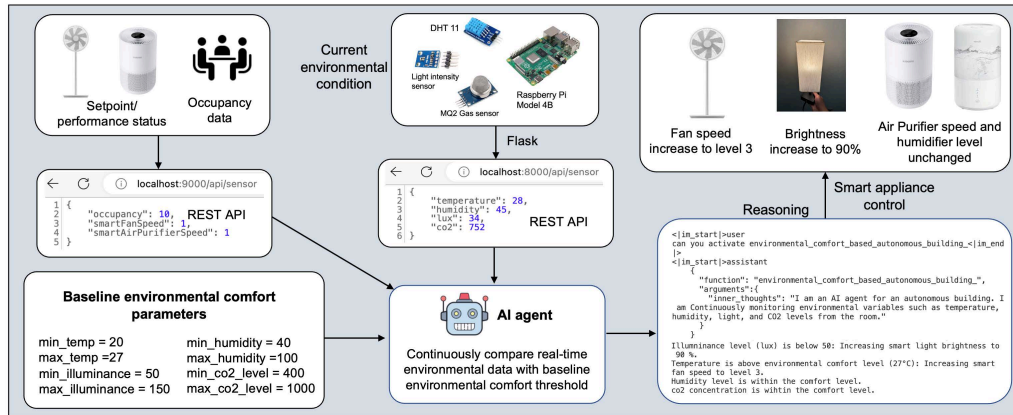


Figure 4: Autonomous smart appliance control using LLM-based AI agent.

and invoke predefined tools or functions to perform tasks. This study leverages llama-cpp-agent, a Python-based package, to implement the function-calling capabilities of the proposed virtual assistant.

5.2 Experiments

The first experiment aims to validate the LLM-based AI agent's ability to control smart building appliances. For this demonstration, the control of a smart light bulb was selected. Users can interact with the virtual assistant to issue voice commands, such as turning the light on or off. The user's prompts are then transmitted to the backend, where they are processed by the locally hosted LLMs. Upon receiving the request, the AI uses its function-calling capabilities to understand and act upon the user's instructions. It then activates a Python script that interfaces with the device's API, enabling it to control the smart light bulb as requested. In addition, instead of the simple on/off commands, users can also provide the context or hint, such as stating that the room is "too dark" or "too bright.". The virtual assistant, leveraging its context-awareness capabilities, can autonomously adjust the brightness level of the light bulb, which can be seen in Figure 3. This dynamic interaction demonstrates the enhanced functionality of the LLM-powered virtual assistant, distinguishing it from traditional AI systems that lack such nuanced environmental awareness.

The second experiment explores the role of the AI agent in managing smart devices for autonomous building operations

through two distinct scenarios: automatic appliance control based on occupancy and autonomous adjustments according to environmental data. In the first case, this experiment uses the simulated occupancy data (randomly selected between 1 and 20) and is updated every 1 minute to test the AI agent's ability to autonomously control devices such as a smart fan and air purifier. For demonstration, this study defines low occupancy as fewer than five individuals, medium occupancy as five to nine individuals, and high occupancy as ten or more individuals. As demonstrated in Figure 4, the AI agent continuously monitors the fan and air purifier settings and occupancy data through RESTful API. Depending on the occupancy levels, the AI modifies the performance of the devices accordingly. If no occupants are detected, the AI will turn off all devices. In the case of low occupancy, it reduces the performance to a lower setting, while in the medium occupancy case, it sets the smart appliance to medium settings. Finally, in high occupancy scenarios, it increases the devices' performance to the maximum setting. Initially, both the smart fan and air purifier were set to low performance at level 1. During the experiment, the simulated occupancy was ten individuals, which led the AI to perform the contextual reasoning and raise the performance settings of the fan and air purifier to level 3 and level 7, respectively, to ensure environmental comfort in the physical space.

The second case aims to validate the AI agent's ability to autonomously adjust smart appliances based on baseline environmental comfort parameters. In this scenario, the smart appliance setpoints were initially configured at their lowest levels.

As shown in Figure 4, the AI retrieved the baseline comfort parameters, which include temperature (20°C-27°C), humidity (40%-100%), light intensity (50-150 lux), and carbon monoxide levels (0-50 ppm). The real-time room conditions of 28°C, 45% humidity, 34 lux, and 752 ppm were obtained through REST API. Upon processing this data, the AI agent determined that the temperature and luminance were outside the comfort range and automatically adjusted the smart fan to speed level 3 and the smart light to 90% brightness while leaving other appliance settings unchanged. This experiment demonstrates the AI agent's ability to autonomously regulate the indoor environment by adjusting smart devices based on real-time data and predefined comfort thresholds. The code for the technical implementation of the prototypes is available under an open-source license [26][27].

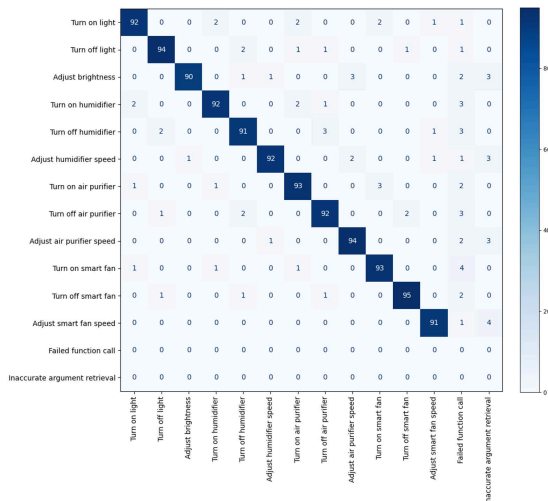


Figure 5: Confusion Matrix for Building Operations Using Virtual Assistant.

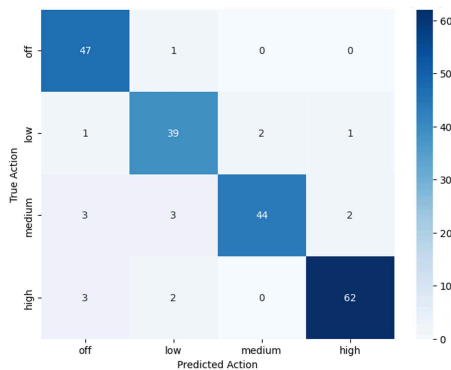


Figure 6: Confusion Matrix for Occupancy-based Autonomous Building Operations.

6 Evaluation

The LLM-based AI systems in this study are composed of two components: a Virtual assistant and an AI agent. The virtual assistants can take user commands and perform smart building tasks by activating tools using function calls. Additionally, the AI agent will monitor the environmental condition occupancy level

and the performance of smart building facilities control using function calls. The performance of the AI systems will be evaluated on the speed, accuracy, and reliability of the function-calling capability of the AI systems. To evaluate the accuracy of the functions or tasks executed by the AI systems, this study uses five different metrics, including Precision, Recall, F1 Score, Overall Accuracy, and Reliability. These metrics are specifically useful for evaluating the model's performance when it successfully selects a function that is similar to the multiclass classification task. The reliability metric aims to examine how consistent the AI is in executing the function when responding to the same prompt.

6.1 Evaluation of the Virtual Assistant

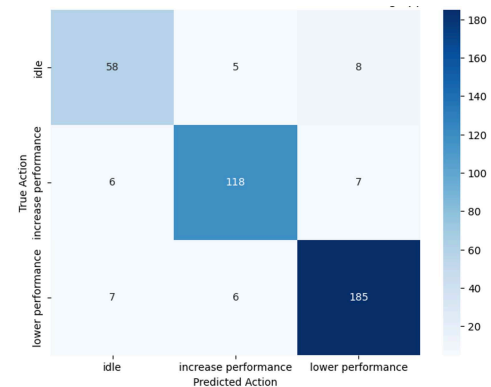


Figure 7: Confusion Matrix for Threshold-based Autonomous Building Operations.

There are 12 different functions/tools that will be used by the virtual assistant upon its reasoning of the user command. Those functions are Turning on the light, turning off the light, adjusting brightness, turning on and off the humidifier, Adjust the humidifier speed, turning on the air purifier, turning off the air purifier, Adjust air purifier speed, Turning on the smart fan, Turning off the smart fan, Adjust smart fan speed. Among the 12 functions, 4 of them, which are Adjust Brightness, Adjust Humidifier Speed, Adjust Air Purifier Speed, and Adjust Smart Fan Speed, will retrieve arguments (e.g., level of fan speed, etc.) from the user command. To demonstrate the variety of user requests, 10 different request prompts were used for each of the 12 functions. For the reliability metric, this study will ask the virtual assistant 10 times for the same prompt within each function. The confusion matrix for building operations with a virtual assistant for each function is shown in Figure 5. It should be noted that there are two additional columns for failed function call and Inaccurate argument retrieval, which aim to demonstrate the times when the AI systems can't understand the command of the user, error, or incorrect retrieval of argument. Additionally, to measure the task execution speed of the AI systems, this study will use tools like Llamacpp to measure the throughput and times needed to execute different tasks.

6.2 Evaluation of the AI Agent

To evaluate the AI agent's performance in automating smart building operations, two experimental setups are used: occupancy-based automation and threshold-based automation. For occupancy-

AI Systems	Task	Precision (%)	Recall (%)	F1 Score (%)	Accuracy (%)	Reliability (%)
AI Assistants	Smart appliance control task	95.91	92.75	94.17	92.75	95.45
AI Agents	Occupancy based automation	90.74	90.23	90.29	91	N/A
	Threshold based automation	88.58	88.16	88.25	90.25	N/A

Figure 8: Evaluation of the proposed AI systems.

based control, we define a predefined range of occupants (x_{true}) (e.g., 0-20 people) and corresponding smart appliance setpoints (y_{true}) (off, low, medium, high) as mentioned in section 5.2. The AI agent is tested with 200 randomized occupancy data points, and its predicted appliance adjustments ($\hat{y}$) are compared against ground truth values (y_{true}). If the AI's assigned setpoint deviates from the predefined range, the prediction is considered incorrect. The accuracy of the system is assessed using precision, recall, and F1-score. The confusion matrix for occupancy-based autonomous building operation with the AI agent for different scenarios is shown in Figure 6.

In addition, to evaluate the AI agent's performance in threshold-based control mode, the study implements predefined environmental control thresholds, including temperature (20°C-27°C), humidity (40%-100%), light intensity (50-150 lux), and carbon monoxide levels (0-50 ppm). These simplified threshold values are implemented primarily to demonstrate the functionality of the LLM-based automation system and its response mechanisms. This experiment involves feeding 400 environmental data points via REST API, where the AI agent continuously monitors real-time temperature, humidity, light intensity, and carbon monoxide levels. If the real-time data falls below or exceeds the predefined threshold, the AI will adjust the corresponding smart appliance by increasing or decreasing its performance to restore the environmental conditions to the comfort range. The AI's adjustments are then validated against predefined expected actions, which are whether to increase or decrease the performance of smart appliances. The confusion matrix for threshold-based autonomous building operation with the AI agent for different scenarios is shown in Figure 7. Accuracy is assessed using precision, recall, F1-score, and macro-averaged performance across all parameters to evaluate the AI's effectiveness in maintaining optimal building conditions. The evaluation dataset including testing prompts for the AI assistant and AI agents is available at [28].

7 Result and Discussion

The evaluation results in Figure 8 show the performance of the proposed virtual assistants and AI agents across different metrics. The virtual assistant for smart appliance control performed exceptionally well, achieving a precision of 95.91%, a recall of 92.75%, and an F1 score of 94.17%, demonstrating high accuracy and reliability in executing user commands. With an accuracy of 92.75% and a reliability score of 95.45%, it proved to be highly reliable in consistent task execution. For the AI agent, the occupancy-based automation system possesses a precision of 90.74%, a recall of 90.23%, and an F1 score of 90.29%, with an accuracy of 91%. These results indicate strong performance in adjusting smart appliances based on occupancy levels. The

threshold-based automation system had slightly lower scores, with precision at 88.58%, recall at 88.16%, and an F1 score of 88.25%, achieving an accuracy of 90.25%.

Furthermore, the scalability of the proposed LLM-based AI systems is evaluated based on its throughput and ability to handle concurrent user requests, specifically measuring how many requests the AI can process simultaneously and how quickly the system can respond to user queries. In this experiment, LlamaBench [29], an open-source tool for benchmarking LLM, is used to assess the performance of the proposed AI-based agent and virtual assistant. The results indicated that execution time and throughput varied based on the specific task. For chat or text generation, the average throughput was 33.66 tokens per second. One token is approximately equivalent to 4 English characters, and 1,500 words correspond to around 2048 tokens [30]. Smart home control tasks took longer, with an average execution time of 5402.62 milliseconds per task and a throughput of 12.77 tokens per second. Concurrency user request is also an important indicator of the LLM model's scalability [31]. This study used Llamacpp for model deployment, which allows parallelization based on the model context length. For instance, a model with a context length of 8192 tokens can theoretically handle 16 parallel requests where each prompt has 256 tokens, and each response generates 256 tokens [32]. Although the Llama 3 model we used supports a context length of up to 128k tokens, we limited it to 8192 tokens due to limited computational resources. For real-world deployment, especially with a larger user base, we can improve the scalability by opting for models with larger context lengths and running them on machines with greater GPU RAM capacity.

8 Conclusion and Future Works

The proposed LLM-powered virtual assistant in this study allows users to interact with the building through voice and text interfaces for smart appliance control. The AI agent also powers the autonomous building operations by autonomously adjusting smart appliances, such as lighting and HVAC, based on occupancy and the baseline environmental comfort threshold to maintain optimal conditions for occupants. The virtual assistant and AI agent demonstrated strong performance with over 90% accuracy, recall, precision, F1, and reliability. In addition, the current trend towards smaller and more efficient language models, as evidenced by Microsoft's Phi-3, Meta's LLaMA 3.2, and Google's Gemma-2 model, indicates that powerful AI systems will be able to effectively deploy on low-cost edge devices such as Raspberry Pi while offering impressive performance. This development could significantly contribute to extending the reach of smart building systems to entire smart cities, enabling more distributed and responsive urban management.

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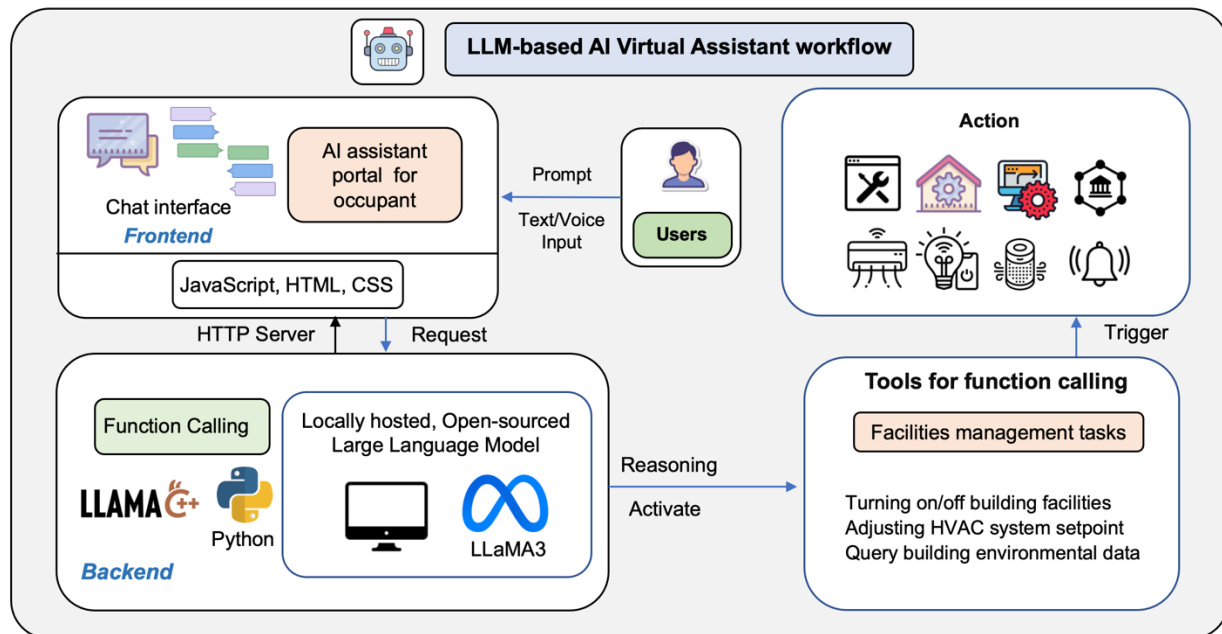
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Project List

1. AI assistant for smart building using Large Language Models

Project description: The Project aims to facilitate the human-building interaction within smart buildings using open-sourced LLM such as LLaMA 3. This AI assistant provides smart and personalized assistance to occupants through web apps. Users can communicate with the AI virtual assistant through text and voice input to control various building facilities, adjust setpoints for the specific building smart facilities, or turn systems on or off as needed. The assistant also provides real-time information on indoor environmental conditions by accessing live sensor data from the IoT device. The Text-to-Speech (TTS) and Speech-to-Text (STT) models are powered by open-source tools and models such as Whisper and Piper.

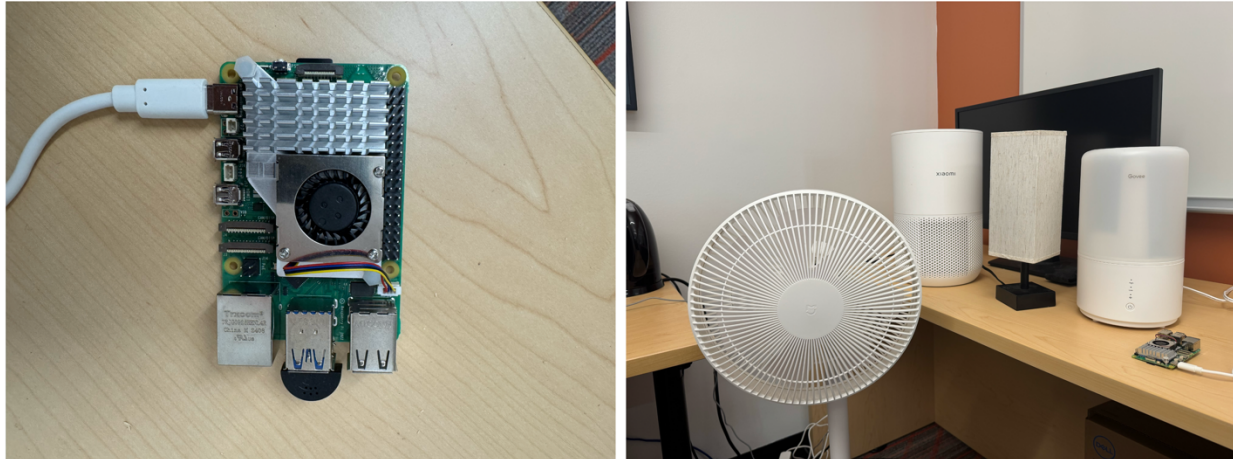
Link: [GitHub Page \(Description, Code, Video demo\)](#)



2. Small Language Model-Based Smart Home Devices for Human-Building Interaction

Project description: This project explores the integration of small language models (SLMs) with smart home environments by deploying the Phi-3 Mini model and Llama 3.2 onto a Raspberry Pi 5 to enable natural language interaction with building systems. The primary objective is to enhance human-building interaction through accessible, cost-effective, and localized AI-powered solutions. This project seeks to revolutionize human-building interaction by embedding generative AI within the built environment. By enabling localized AI-driven decision-making, the research contributes to the development of more responsive, secure, and sustainable smart home ecosystems, forming the backbone for future smart cities.

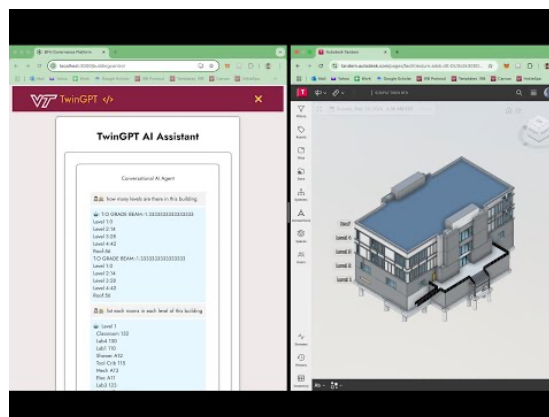
Link: [GitHub Page \(Description, Code, Video demo\)](#)



3. TwinGPT: Large Language Model-based AI assistant for Digital twin query

Project description: TwinGPT is an AI-powered digital twin query system designed for facility managers to interact with and extract insights from building data using a large language model (LLM). This system integrates static data, such as BIM (Building Information Modeling) details such as Rooms, Levels, and building assets, with dynamic data from sensors, environmental inputs, and occupancy information. It provides an intuitive interface for querying both historical and real-time building data. The platform offers an interactive AI chatbot, enabling facility managers to ask questions or communicate with the AI assistant to retrieve relevant building information quickly and efficiently. TwinGPT supports queries that span static BIM data and dynamic inputs, such as sensor readings, environmental conditions, and occupancy metrics. This platform is powered by the open-source AI system Llama 3.2, powered by the llama.cpp framework.

Link: [GitHub Page \(Description, Code, Video demo\)](#)



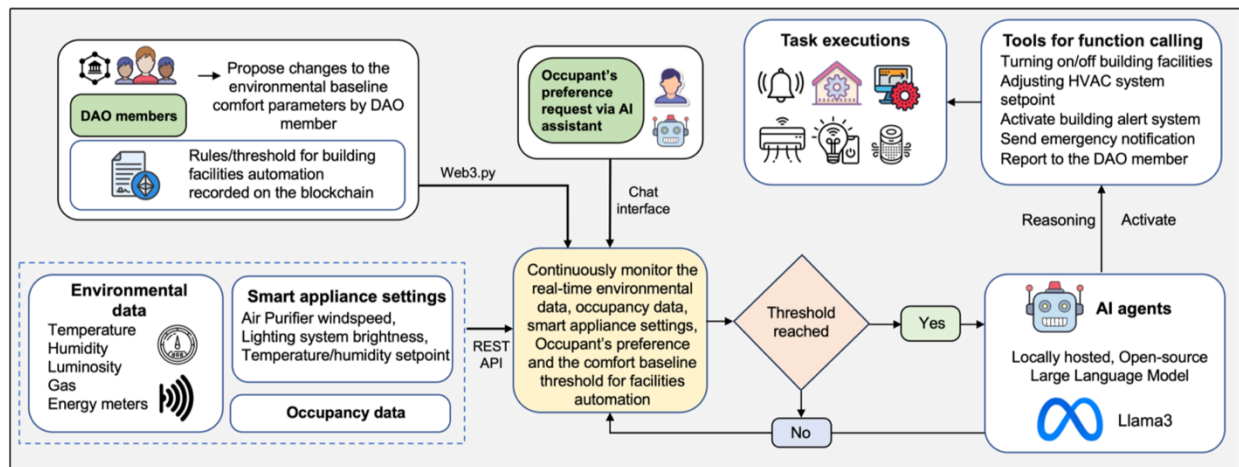
4. AI agent for Autonomous Building operation using Large Language Models

Project description: This project introduces an advanced AI agent designed to oversee autonomous building operation and facilities management. The proposed LLM-based autonomous building operation framework combines IoT devices, smart building facilities, LLMs, and blockchain technologies to create an automated system for managing building systems.

Threshold-Based Automation System: The threshold-based automation system utilizes the AI agent to continuously monitor real-time sensor data and compare it with these predefined thresholds. If the environmental conditions exceed these thresholds, the AI agent uses its function-calling capabilities to take action, such as adjusting HVAC setpoints, modifying lighting intensity, or triggering alerts to maintain optimal building conditions. These threshold values are the baseline comfort parameters for ensuring optimal comfort and indoor environmental quality.

Occupancy-Based Automation System The proposed AI agent will adjust the performance of building systems (e.g., HVAC performance or lighting intensity) based on occupancy level to dynamically optimize energy consumption. For example, during low occupancy periods, the AI agent may reduce HVAC intensity or dim the lighting to conserve energy. Conversely, during high occupancy, the agent ensures that environmental conditions are optimal for occupants. The system also accommodates user preferences and feedback through natural language processing, enabling users to provide specific requests or adjust system behaviors dynamically.

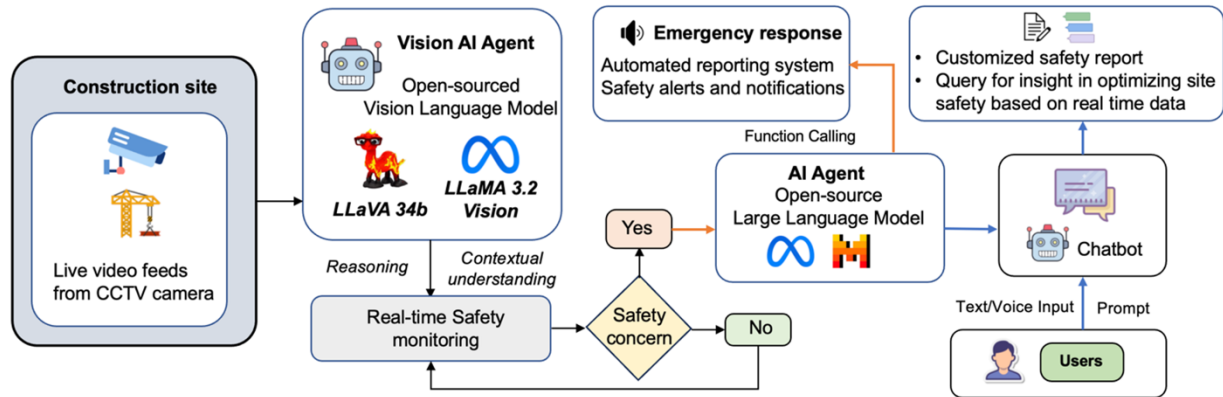
Link: [GitHub Page \(Description, Code\)](#)



5. Vision Language Model-based AI agent for Construction Site Progress and Safety Monitoring

Project description: This project involves developing a vision-language model (VLM)-based autonomous system to monitor construction site progress and safety conditions, thereby providing a comprehensive analysis of construction sites to ensure ongoing progress and adherence to safety protocols.

Link: [GitHub Page \(Description, Code, Video demo\)](#)



6. Vision Language Model-based AI assistant and AI agents and (XR) Extended Reality application

Project description: This project explores the integration of Vision Language Model (VLM), Large Language Model (LLM), and extended reality (XR) to create a Multimodal AI assistant with voice chat, Image understanding, and smart building control in immersive environments. This project aims to create an innovative solution for remote facility management and urban infrastructure monitoring.

The developed system deploys an LLM-based AI assistant and a digital building twin into an XR environment using Microsoft HoloLens 2. Users can interact with the BIM models and communicate with the Multimodal AI chatbot. Users can also interact with the AI assistant through voice commands to control building facilities. This setup enhances the ability of facility managers and occupants to interact with and control smart buildings remotely. The approach also holds the potential for scaling to multiple buildings or urban infrastructure, enabling immersive, real-time monitoring and management for smart city applications.

Link: [GitHub Page \(Description, Code, Video demo\)](#)



(a) AI Virtual Assistant

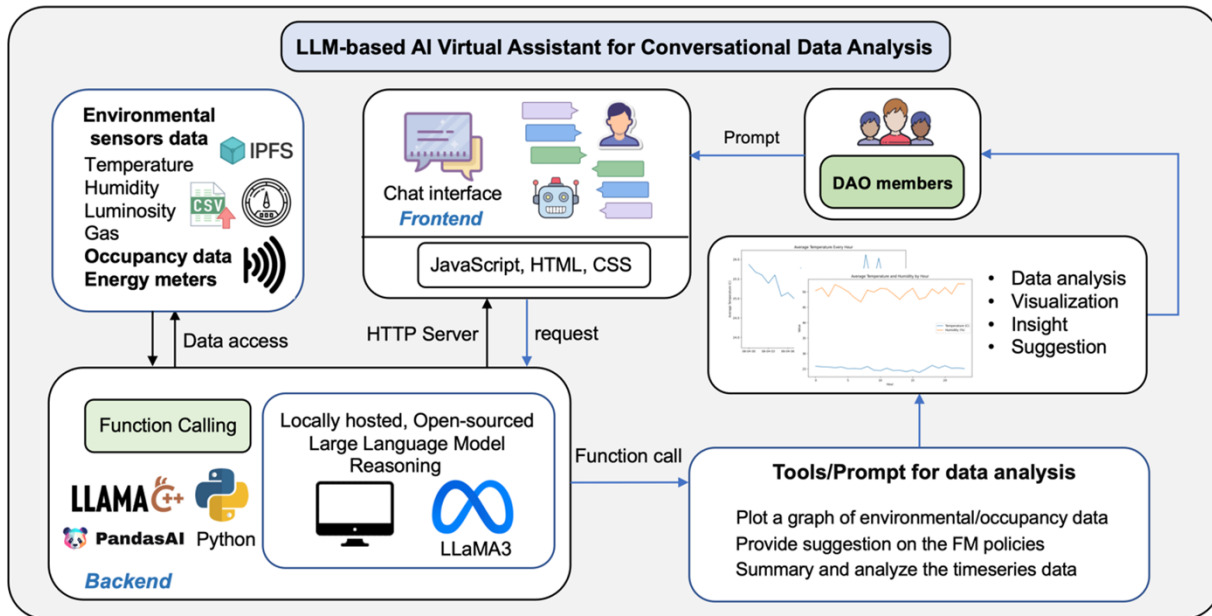


(b) Augmented digital building twin with AI assistant

7. Data-driven Smart Building Facilities Management using AI Assistant and Digital Twin

Project description: This project focuses on implementing an AI and Digital twin-driven decision support system for smart building facilities management using digital twins, and large language models (LLMs). The aim is to enhance facility management through AI-driven insights and digital twin visualizations.

Link: [GitHub Page \(Description, Code, Video demo\)](#)



8. Retrieval-Augmented Generation (RAG) Chatbot for Construction Safety using LLM

Project Description: This project developed a RAG chatbot using LLaMA 3, Llamaindex, and a vector database to provide construction personnel with instant access to safety protocols and building codes. The AI-powered chatbot offers accurate and contextually relevant information from a construction safety standards database. This system enhances safety awareness and promotes a safer construction environment through efficient knowledge retrieval.

Link: [GitHub Page \(Description, Code, Video demo\)](#)

9. A GenAI-Powered Prototype for Detecting Defects and Vulnerabilities in Smart Home Automation Systems

Project description: To address this challenge, this project will develop SmartHomeSecure, an intelligent software prototype leveraging advanced Generative AI, LLaMA, to enhance smart home security and reliability. Specifically, the prototype will (i) identify and correct errors in YAML-based automation configurations through syntax and semantic analysis, (ii) detect vulnerabilities in smart home automations using control flow analysis, symbolic execution, and GenAI-powered insights, and (iii) simulate real-world scenarios to test automation performance under diverse conditions.

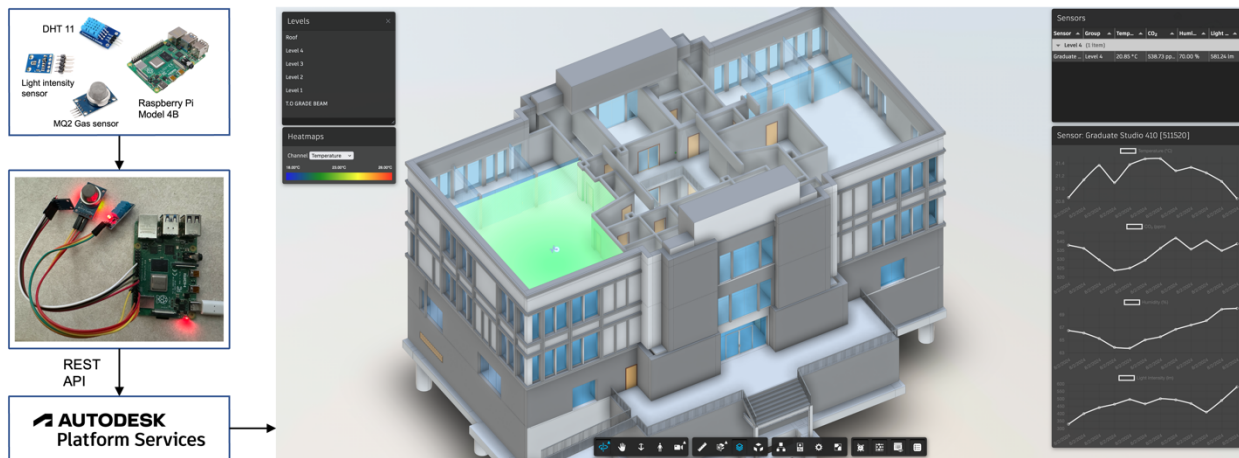
Link: [GitHub Page \(Description, Code, Video demo\)](#)

10. Digital building twin

Project description:

This project focuses on developing digital twins for visualization of environmental conditions within a physical building. The framework is demonstrated using a case study of Bishop-Favrao Hall, the home of the Department of Building Construction at Virginia Tech. This digital twin provides real-time visualization of environmental conditions, including temperature, humidity, lighting levels, and air quality, within the building.

Link: [GitHub Page \(Description, Code, Video demo\)](#)



11. Decentralized Digital building twin using Public and Private Blockchain network

Project description: This project focuses on developing a blockchain-based IoT framework and digital twin model to automate and optimize building facilities operations. The project leverages Hyperledger Fabric, Ethereum blockchain, digital twin technologies, and various IoT sensors and devices.

Link: [GitHub Page \(Description, Code, Video demo\)](#)

